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MEASUREMENT AND DETERMINATION OF CRYSTALLOGRAPHIC TEXTURE WITH RESPECT TO POSITION

Abstract

An example method includes measuring, by at least one of a polarized light device, a spatially resolved acoustic spectroscopy device, or an eddy current device, an alpha phase data set indicative of an alpha phase of a crystalline structure of a material. The method includes receiving, by processing circuitry, the alpha phase data set, wherein the alpha phase data set comprises a plurality of pixels, wherein each pixel of the plurality of pixels includes a position, a first Euler angle, a second Euler angle, and a third Euler angle, wherein the third Euler angle is missing or erroneous. The method also includes adjusting, by the processing circuitry, the third Euler angle of a pixel of the plurality of pixels and storing, by the processing circuitry and based on adjusting the third Euler angle of the pixel reducing a total beta phase misorientation, the alpha phase data set.

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Background/Summary

TECHNICAL FIELD

[0002] This disclosure relates to determining the local crystallographic texture of a polycrystalline material.

BACKGROUND

[0003] Components of high-temperature mechanical systems, such as gas-turbine engines, operate in severe environments. Some components may be formed of a metal or metal alloy, such as, for example, titanium or a titanium alloy. Other components may be formed of a ceramic or a composite material.

[0004] Mechanical properties of a material may depend at least in part on the microstructure of the material, including a presence or absence of defects, such as holes, voids or sections with a different chemical composition or phase constitution, within the material. For this reason, knowledge of the presence or absence of defects in the material may be desired before utilizing the material in a component, such as a gas turbine engine component.

SUMMARY

[0005] In general, this disclosure describes techniques for determining a crystalline structure of a material based on measurements including missing and/or erroneous data. More particularly, this disclosure describes techniques and devices for determining third Euler angles with improved accuracy based on measurement data sets including first and second Euler angles and missing and/or erroneous third Euler angles.

[0006] In one example, this disclosure describes a method including: measuring, by at least one of a polarized light device, a spatially resolved acoustic spectroscopy device, or an eddy current device, an alpha phase data set indicative of an alpha phase of a crystalline structure of a material; receiving, by processing circuitry, the alpha phase data set, wherein the alpha phase data set comprises a plurality of pixels, wherein each pixel of the plurality of pixels includes a position, a first Euler angle (ϕ_1), a second Euler angle (Φ), and a third Euler angle (ϕ_2), wherein the third Euler angle (ϕ_2), is missing or erroneous; adjusting, by the processing circuitry, the third Euler angle (ϕ_2), of a pixel of the plurality of pixels; and storing, by the processing circuitry and based on adjusting the third Euler angle (ϕ_2), of the pixel reducing a total beta phase misorientation, the alpha phase data set.

[0007] In another example, this disclosure describes a device including: a measurement device configured to measure a crystalline structure of a material and output an alpha phase data set indicative of the measurement of the crystalline structure; a memory; and one or more processors implemented in circuitry and in communication with the memory, the one or more processors configured to: receive the alpha phase data set, wherein the alpha phase data set comprises a plurality of pixels, wherein each pixel of the plurality of pixels includes a position, a first Euler angle (ϕ_1), a second Euler angle (Φ), and a third Euler angle (ϕ_2), wherein the third Euler angle (ϕ_2), is missing or erroneous; adjust the third Euler angle (ϕ_2), of a pixel of the plurality of pixels; and store, and based on adjusting the third Euler angle (ϕ_2) of the pixel reducing a total beta phase misorientation, the alpha phase data set.

[0008] In another example, this disclosure describes a non-transitory computer-readable storage medium having stored thereon instructions that, when executed, configure a processor to: receive an alpha phase data set, wherein the alpha phase data set comprises a plurality of pixels, wherein each pixel of the plurality of pixels includes a position, a first Euler angle (ϕ_1), a second Euler angle (Φ), and a third Euler angle (ϕ_2), wherein the third Euler angle (ϕ_2) is missing or erroneous; adjust the third Euler angle (ϕ_2) of a pixel of the plurality of pixels; and store, based on adjusting the third Euler angle (ϕ_2) of the pixel reducing a total beta phase misorientation, the alpha phase

data set.

[0009] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

[0011] As the color drawings are being filed electronically, only one set of the drawings is submitted.

[0012] FIG. 1 is a block diagram illustrating an example of a system that may be used to determine microtexture regions (MTRs) and MTR size statistics within a sample.

[0013] FIG. 2 is a functional block diagram illustrating another example of a system which may be used to determine MTRs and MTR size statistics within sample.

[0014] FIG. 3 is a schematic representation of an example transformation between a beta phase grain and a plurality of alpha phase grains of a material.

[0015] FIG. 4 is a schematic illustration of an example beta phase lattice and an example alpha phase lattice related through a Burgers orientation relationship.

[0016] FIG. 5A is a plot of an example total beta phase misorientation curve as a function of iteration number of an iterative algorithm for determining third Euler angles from a data set of measured orientations of a material.

[0017] FIG. 5B is a schematic illustration of an example measurement area including a plurality of alpha phase grains after transforming from an example plurality of prior beta grains.

[0018] FIG. 6A is a schematic diagram illustrating an example measurement area of a measurement device.

[0019] FIG. 6B is a schematic diagram of an example database module configured to store a plurality of values, data sets, data, and/or information.

[0020] FIG. 7 is a flow diagram of an example method of determining third Euler angles of a data set measured by a measurement device.

[0021] FIG. 8 is a flow diagram of an example method of determining which alpha phase grains may be from the same prior beta grain.

[0022] FIG. 9 is a flow diagram of an example method of adjusting the value of third Euler angles, determining a beta phase data set, and determining a Burgers inverse transformation solution that reduces a total beta phase misorientation.

[0023] FIG. 10 is a schematic diagram illustrating the six unique Burgers IT solutions.

[0024] FIG. 11 is a flow diagram of an example method of refining selection of the first three Burgers IT solutions.

[0025] FIG. 12 is a flow diagram of an example method of refining selection of the second three Burgers IT solutions.

[0026] FIG. 13 is a flow diagram of an example method of refining selection of all six of Burgers IT solutions.

[0027] FIG. 14 is a flow diagram of an example method of refining values of third Euler angles as well as selection of all six Burgers IT solutions.

[0028] FIG. 15 is a flow diagram of an example method of further refining values of third Euler angles as well as selection of all six Burgers IT solutions.

[0029] FIG. 16 is a flow diagram of an example method of further refining values of third Euler

angles as well as selection of all six Burgers IT solutions.

[0030] FIG. **17** illustrates an example rotation matrix which may be used to determine the crystal axes and specimen directions of a crystal relative to the specimen coordinate system.

[0031] FIG. **18** is a set of two-dimensional heat maps illustrating the results an example alpha phase data set determined via the method of FIG. **7** compared with an electron backscatter diffraction measurement of the alpha phase data set, each for an area including the same single prior beta grain.

[0032] FIG. **19** is a table of the misorientations and axis direction for alpha phase orientations from a single prior beta grain corresponding to the alpha phase data set of FIG. **18**.

[0033] FIG. **20** is an illustration of an example correspondence of a deduced (110).sub. β pole figure with (0001).sub. α third Euler angle deduced and measured pole figures.

[0034] FIG. **21** is an illustration of an example correspondence of the deduced (111).sub. β pole figure with the (11-20).sub. α third Euler angle deduced and measured pole figures.

[0035] FIG. **22** is a set of two-dimensional heat maps illustrating the results an example alpha phase data set determined via the method of FIG. **7** compared with an electron backscatter diffraction measurement of the alpha phase data set, each for an area including the same plurality of prior beta grains.

[0036] FIG. **23** is an illustration of plots of the alpha and beta phase orientations for two adjacent pixels of an example alpha data set having a c-axis misorientation of 0° or 1800.

[0037] FIG. **24** is an illustration of plots of the alpha and beta phase orientations for two adjacent pixels of an example alpha data set having a c-axis misorientation of 300.

[0038] FIG. **25** is an illustration of plots of the alpha and beta phase orientations for two adjacent pixels of an example alpha data set having a c-axis misorientation of 60°.

[0039] FIG. **26** is an illustration of plots of possible 300 third Euler angle misorientations due to cubic and hexagonal symmetries in relation to a cartesian coordinate system.

[0040] FIG. **27** is a table of example cubic symmetry operators.

[0041] FIG. **28** is a table of example hexagonal symmetry operators.

[0042] FIG. **29** is a plot comparing MTR size distribution of the same area of a material using EBSD versus other measurement techniques including missing and/or inaccurate third Euler angles.

DETAILED DESCRIPTION

[0043] In general, the present disclosure is directed to systems, devices, and techniques for determining a structure, a crystalline structure, and/or a defect in a material and, optionally, determining an approximate size and/or shape of the crystalline structure and/or defect. The systems, devices, and techniques may determine which areas of a material include microtexture regions (MTRs) and may determine a crystalline orientation of MTRs and non-MTR areas.

[0044] MTRs are large regions of the microstructure in forged components that have the same crystal orientation. MTRs may originate from a billet used in a forging process and may flow like rivers along the plastic flow lines of the forging and may then be broken up into separate regions or areas. Characterizing MTRs in certain zirconium and titanium alloys, such as Ti-6Al-4V, is of interest in many industries, including the aerospace industry.

[0045] MTRs may be determined, measured, and/or characterized using electron backscatter diffraction (EBSD) via a scanning electron microscope (SEM). EBSD characterization produces a data set (e.g., a “data file”) containing the two-dimensional positions, e.g., x and y coordinates, of a measurement location (which may be referred to as a “pixel”) along with three Euler angles (ϕ_1 , Φ , and ϕ_2) that describe the crystal orientation at the location “x” and “y” in relation to a reference, or specimen, coordinate system.

[0046] An EBSD data set including three Euler angles at each of a plurality of (x, y) positions may be used to determine crystal orientation, and misorientations relative to other positions, and to cluster or segment the EBSD data set to describe a distribution of MTRs and MTR size statistics present in a forged product (e.g., mean, median, max and minimum MTR sizes). However, a

challenge with using EBSD to measure and characterize MTRs is that MTRs in forged products may be larger than the practical measurement field of view of a SEM. For example, MTRs may span tens of millimeters (mm) in length in a billet product, whereas SEM may have a field of view that is much smaller than tens of millimeters. SEM may measure larger areas by translating the sample (or detectors) and stitching multiple measurements together, however, SEM is performed in a vacuum and SEM measurement areas are typically limited by the amount of translation available within the vacuum chamber, making SEM measurement of larger areas more difficult, time-consuming, and costly, if not impossible due to translation limitations.

[0047] For this reason, alternative measurement and characterization devices and techniques have been developed, for example, polarized light devices and techniques, spatially resolved acoustic spectroscopy (SRAS), and eddy current devices and techniques. However, alternative measurement and characterization devices and techniques do not have the ability to accurately determine the third Euler angle, e.g., ϕ_2 . As such, the alternative measurement and characterization devices and techniques are not able to differentiate at least certain orientations of crystals, which may preclude accurate determination of MTRs and MTR statistics. For example, alternative measurement and characterization devices and techniques cannot distinguish between two crystals that have their c-axis perfectly aligned, but are rotated about their c-axis, because of inaccurate measurements of the third Euler angle ϕ_2 . Additionally, for more complicated mis-orientations (e.g., between two crystals) the contribution of the third Euler angle ϕ_2 on the mis-orientation cannot be determined. FIG. 29 is a plot comparing MTR size distribution of the same area of a material using EBSD versus other measurement techniques including missing and/or inaccurate third Euler angles.

[0048] In accordance with the systems, devices, and techniques described herein, a method of determining MTRs and MTR size statistics, via determining the third Euler angle (ϕ_2) from data sets of alternative measurements (e.g., alternative to EBSD) having inaccurate, or missing, third Euler angle (ϕ_2) data, includes measuring an alpha phase data set indicative of an alpha phase of a crystalline structure of a material, such as Ti-6Al-4V. The measurement device may be a device that outputs the alpha phase data set with missing and/or erroneous third Euler angles. Processing circuitry, such as an analysis module, may be configured to receive the alpha phase data set, the alpha phase data set including a plurality of pixels, each pixel including a position, first Euler angles (ϕ_1), second Euler angles ϕ_2 , and missing and/or erroneous third Euler angles (ϕ_2). The processing circuitry may be configured to adjust the third Euler angle (ϕ_2) of the pixels, e.g., to correct the third Euler angle (ϕ_2) of the pixels, and store the alpha phase data set, e.g., for further processing. In some examples, the processing circuitry may be configured to determine, subsequent to changing the value of the third Euler angle (ϕ_2) of the pixels, at least one microtexture region (MTR) based on the alpha phase data set, output an alert based on determining the at least one MTR, determine a status of a component made of the material, e.g., a blade of a gas turbine engine, based on at least one of the alpha phase data set or the at least one MTR, and output the status of the blade, e.g., to another device and/or a display.

[0049] Example devices, systems, and methods described herein may provide several benefits and/or advantages. For example, the systems and techniques may improve the accuracy of measurements of materials with areas large enough to include MTRs, the improvement in accuracy enabling determination of the presence and size statistics of MTRs based on data sets having a large enough field of view and/or measurement area but including missing or erroneous data. The systems and techniques may provide lower cost alternatives, e.g., relative to EBSD, which may increase measurement and characterization of potential defects and improve performance of devices utilizing such materials.

[0050] FIG. 1 is a block diagram illustrating an example of a system 10 that may be used to determine MTRs and MTR size statistics within sample 16. System 10 includes a data analysis device 12 and measurement device 14. Measurement device 14 is configured to measure a first surface 22 of sample 16.

[0051] Sample **16** may be any material, including, for example, a ceramic, a metal or a metal alloy, or a composite material. For example, sample **16** may include an alloy used in a high temperature mechanical system, such as a blade, or other component, of a gas turbine engine. Such alloys include titanium-based, nickel-based, magnesium-based or zirconium-based superalloys, e.g., Ti-6Al-4V, Ti-6242, Ti-17, Ti-6Al, Zr-2.5% Nb, Zircalloy, or the like.

[0052] System **10** may be utilized to detect the presence of one or more MTRs and MTR size statistics of the one or more MTRs. In some examples, system **10** may determine whether the one or more MTRs may be characterized as a defect or defects within sample **16**. In some examples, system **10** may allow determination of the approximate size and/or shape of one or more MTRs, and may additionally or alternatively facilitate determination of locations of the one or more MTRs within sample **16**. System **10** includes data analysis device **12**, which controls operation of system **10** automatically or under control of a user **42** (FIGS. 2 and 3).

[0053] Data analysis device **12** may be a general-purpose workstation, desktop computer, laptop computer, a handheld computing device, a personal digital assistant (PDA), or other computing device. Data analysis device **12** may include a microprocessor, digital signal processor (DSP), field programmable gate array (FPGA), application specific integrated circuit (ASIC) or other hardware, firmware and/or software for implementing the techniques described in this disclosure. In other words, the control of system **10** and analysis of data sets, as described herein, may be implemented in hardware, software, firmware, combinations thereof, or the like. If implemented in software, a computer-readable medium may store instructions, i.e., program code, that can be executed by a processor or DSP to carry out one or more of the techniques described herein. For example, the computer-readable medium may comprise magnetic media, optical media, random access memory (RAM), read-only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), magnetoresistive random access memory (MRAM), flash memory, or other media suitable for storing program code.

[0054] Data analysis device **12** controls operation of measurement device **14** and receives from measurement device **14** signals representative of the positions and Euler angles ($\phi 1$, Φ , and $\phi 2$) measured and/or detected by measurement device **14**.

[0055] Measurement device **14** may be configured to measure a two-dimensional area of sample **16** at surface **22** and output a data set, or “image” of the structure sample **16**, e.g., the crystalline structure and crystalline orientation of sample **16**. Measurement device **14** may output an “image” where each x-y location, or “pixel,” of the image includes a set of the three Euler angles ($\phi 1$, Φ , and $\phi 2$) measured by measurement device **14**. Each pixel may represent a crystal, or a portion of a crystal, of sample **16** at the pixel location, e.g., the crystalline structure of sample **16** at the pixel location. In some examples, measurement device **14** may be a polarized light measurement device, a SRAS measurement device, an eddy current measurement device, or the like. In some examples, measurement device **14** may measure, and output, a data set or image of Euler angles ($\phi 1$, Φ , $\phi 2$) in which the third Euler angle, $\phi 2$, is inaccurate, erroneous, or missing. In some examples, measurement device **14** may be configured to measure an area of surface **22** of sample **16** that is relatively small (e.g., 5 mm×5 mm) to something extremely large (1 meter×1 meter, or larger) through the stitching of data from smaller fields of view together. For example, measurement device **14** may not need to be performed under vacuum, and may not be limited by a vacuum chamber, e.g., in translating sample **16** or detectors of measurement device **14**.

[0056] In some examples, one or more processors, or processing circuitry, of data analysis device **12** may be configured to control the measurement device **14** to measure an area of a surface of a material, e.g., area **650** (FIG. 5B) of surface **22** of sample **16** (FIG. 2). The measurement may yield a data set including information indicative of an alpha phase of a hexagonally close packed (hcp) crystalline structure and/or orientation of sample **16**, e.g., Euler angles ($\phi 1$, Φ , and $\phi 2$). The measurement device and/or technique may yield a data set including missing and/or erroneous information, e.g., missing and/or erroneous third Euler angles ($\phi 2$). For example, the measurement

device may be a polarized light device, a spatially resolved acoustic spectroscopy device, or an eddy current device.

[0057] In some examples, the alpha phase data set may include a plurality of pixels, each including information such as the Euler angles as a function of position. The processing circuitry may be configured to receive the alpha phase data set, and to adjust the third Euler angles (ϕ_2) of pixels of the data set, e.g., to correct the third Euler angle (ϕ_2) of the pixels, and store the alpha phase data set, e.g., for further processing. In some examples, the processing circuitry may be configured to determine, subsequent to changing the value of the third Euler angle (ϕ_2) of the pixels, at least one microtexture region (MTR) based on the alpha phase data set, output an alert based on determining the at least one MTR, determine a status of a component made of the material, e.g., a blade of a gas turbine engine, based on at least one of the alpha phase data set or the at least one MTR, and output the status of the blade, e.g., to another device and/or a display.

[0058] FIG. 2 is a functional block diagram illustrating an example of a system **10** which may be used to determine MTRs and MTR size statistics within sample **16**. In the example illustrated in FIG. 2, system **27** includes measurement device **14** and a data analysis device **31**. Data analysis device **31** includes a control module **28**, a communication module **30**, an analysis module **32**, a database module **34**, and an interface module **36**. Data analysis device **31** may be substantially similar to data analysis device **12** of FIG. 1, except for the differences described herein.

[0059] Interface module **36** represents software and hardware necessary for interacting with a user, e.g., for receiving input from a user **42** and for outputting information to the user **42**. Interface module **36** may receive input from input devices **40** and output data to output devices **38** that enable a user **42** to interact with data analysis device **12**. For example, via interface module **36**, user **42** may change operational parameters of data analysis device **12** and manipulate data stored in database module **34**. Moreover, user **42** may interact with interface module **36** to initiate measurement of sample **16** to detect materials properties (e.g., crystal structure and orientations) and determine the presence of, and size statistics of, MTRs. Further, user **42** may interact with data analysis device **12** to view and manipulate the acquired data via output devices **38** and input devices **40**. During this process, interface module **36** may present a user **42** with user interface screens for interacting with analysis device **12**. Exemplary input devices **40** include a keyboard, a touch screen, a mouse, a microphone, and the like. Output devices **38** may include, for example, an LCD screen, an LED array, a CRT screen, or a touch screen display.

[0060] Communication module **30** represents hardware and software necessary for communication between data analysis device **31** and another device, such as, for example, a device external to system **27**, such as another computing device. The communication module **30** may include a single method or combination of methods to transfer data to and from data analysis device **31**. Some methods may include a universal serial bus (USB) port, a PCI bus, or IEEE 1394 port for hardware connectivity with high data transfer rates. In some examples, a storage device may be directly attached to one of these ports for data storage for post processing. The data may be pre-processed by control module **28** and/or analysis module **32** and ready for viewing, or the raw data may need to be completely processed before analyzing can begin.

[0061] Communication module **30** may also include radio frequency (RF) communication or a local area network (LAN) connection. Moreover, communication may be achieved by direct connection or through a network access point, such as a hub or router, which may support wired or wireless communications.

[0062] Control module **28** represents control logic that, in response to input received from user **42** via interface module **36**, directs the operation of data analysis device **31** and measurement device **14**. For example, control module **28** may comprise software instructions that, when executed, provide control logic for communicating commands to measurement device **14** to commence measurement and data collection via measurement device **14**. Furthermore, control module **28** provides control logic for storing the collected data within database module **34**, and for invoking

analysis module **32** to process the data automatically or in response to commands from user **42**.

[0063] Control module **28** may be configured to control measurement device **14** to generate and direct at least a portion of energy to sample **16** and first surface **22**. For example, control module **28** may be configured to control measurement device **14** to generate and direct polarized light to first surface **22** for a polarized light measurement, or to generate and direct acoustic energy to and/or along first surface **22** for an SRAS measurement, or to generate and direct electrical energy to and/or along first surface **22** for an eddy current measurement.

[0064] Control module **28** may be configured to control measurement device **14** to detect and/or measure return energy from sample **16** and/or first surface **22**, e.g., light (electromagnetic) energy, acoustic energy, and/or electrical current after having interacted with sample **16** and/or first surface **22**, and to acquire one or more signals proportional to the detected/measured energy. Control module **28** and/or measurement device **14** may be configured to then generate a data set representative of the acquired signals, the data set representing detected/measured properties of sample **16** and/or first surface **22**, e.g., Euler angles of the crystalline structure of sample **16** as a function of position of an area of first surface **22**.

[0065] In some examples, the measured data sets, or “images,” may be stored in database module **34**. Database module **34** represents hardware and software necessary for storing and retrieving data, and may comprise, for example, a suitable magnetic media, optical media, random access memory (RAM), read-only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), flash memory, or other media suitable for storing program code and data. In some examples, control module **28** causes the data sets to be stored in database module **34** for later manipulation or may communicate the data sets to analysis module **32** for analysis according to one or more techniques described herein.

[0066] Analysis module **32** may be configured to receive the data sets from control module **28**, to process the data according to at least one of the techniques described herein, and to detect a presence of MTRs and size statistics of MTRs, automatically or in response to an instruction received from user **42**. Analysis module **32** represents processing logic, processing circuitry, and/or control circuitry configured to process data sets according to the methods and techniques described herein. For example, analysis module **32** may comprise software instructions that, when executed, provide control logic for processing data sets and executing the methods and techniques described herein.

[0067] FIG. **3** is a schematic representation of a transformation between a beta phase grain **100** and a plurality of alpha phase grains **102** of a material. For example, a material that is heat treated above a beta transus temperature may have a beta phase crystalline structure at temperatures above the beta transus temperature, and the material may undergo a phase transformation to an alpha phase crystalline structure when cooling to a temperature below the beta transus temperature. The beta transus temperature of a material is the lowest temperature at which substantially all of the material is in the beta phase. The beta phase of a titanium alloy such as Ti-6Al-4V has a body-centered cubic (bcc) crystalline structure. The beta transus temperature may range from about 700 degrees Celsius ($^{\circ}\text{C}$.) to about $1,050^{\circ}\text{C}$. The beta transus temperature of Ti-6Al-4V may be about 995°C . to about $1,000^{\circ}\text{C}$.

[0068] Above the beta transus temperature, the material may include a plurality of grains, or crystals, in the beta phase. Each grain comprises a volume of material having the same orientation of beta phase, e.g., a volume of material comprising a single crystal having a beta phase in a particular orientation. Different beta grains may have different orientations, and each beta grain may be referred to as a prior beta grain, e.g., in reference to the beta grain prior to undergoing a phase transformation from the beta phase to an alpha phase upon the material cooling below the beta transus temperature. For titanium alloys, the beta phase includes atoms in a body centered cubic (bcc) lattice, and the alpha phase includes atoms in a hexagonally close packed (hcp) lattice.

[0069] Upon cooling below the beta transus temperature, each prior beta grain of an alloy, e.g., Ti-

6Al-4V, may undergo a phase transformation to a plurality of alpha grains, e.g., a plurality of volumes having different orientations of an hcp lattice. In the example shown, prior beta grain **100** may undergo a phase transformation **110** to alpha phase material **102** including a plurality of alpha grains (alternatively referred to as alpha phase grains), e.g., alpha grains, **104**, **106**, **108**. In some examples, less than all of a prior beta grain may transform to an alpha phase. For example, a portion of a prior beta grain may transform to a plurality of plurality of alpha grains, and a portion may remain in a beta phase. In some examples, about 95% of the material of a prior beta grain may transform, upon cooling below the beta transus temperature, to a plurality of alpha grains and about 5% may remain in the beta phase.

[0070] FIG. **4** is a schematic illustration of a beta phase lattice **200** and an alpha phase lattice **202** related through a Burgers orientation relationship **210**. Alpha and beta phase alloys, e.g., including titanium alloys such as Ti-6Al-4V, develop complex textures during thermomechanical processing due to the allotropic phase transformation between the beta and alpha phases. These phases are crystallographically related through the Burgers orientation relationship **210**. In some examples, an alloy may undergo a phase transformation upon cooling from a beta phase to only a finite set of alpha phase orientations, e.g., a prior beta grain may not phase transform to any arbitrary alpha phase orientation because only a finite set of alpha phase orientations, e.g., variants, are thermodynamically allowed. For example, a body centered cubic prior beta grain **100** of a titanium alloy such as Ti-6Al-4V may transform to only twelve variants of a hexagonally close packed alpha phase.

[0071] The variants of alpha phase are related to the prior beta grain via the Burgers orientation relationship **210** such that the orientation of the prior beta grain **100** may be calculated based on a measured orientation of any one of the 12 alpha phase variants. For example, alpha grains **104**, **106**, and **108** are each a variant of an orientation of a hexagonally close packed crystal having an orientation related to the same prior beta grain **100**. If the orientation of each of alpha grains **104**, **106**, and **108** are measured, e.g., via EBSD (which includes relatively accurate measurements of all three Euler angles), the orientation of prior beta grain **100** may be calculated based on any one of the measurements of alpha grains **104**, **106**, and **108** via Burgers orientation relationship **210**.

Calculation of a prior beta grain orientation from an orientation of an alpha grain and/or calculation of an orientation variant of an alpha grain from an orientation of a prior beta grain may be referred to herein as “transforming” between the prior beta grain and the alpha grains. If transformation to a prior beta grain from two different alpha grains results in different orientations of the prior beta grain, the two alpha grains did not originate from the same prior beta grain. Two different alpha grains may transform to the same prior beta grain with slightly different orientations, e.g., there may be small variations in the orientation calculated via Burgers orientation relationship **210** due to noise in measurements of the alpha phase variants.

[0072] Burgers orientation relationship **210** may be used to determine which data points (e.g., x-y positions) of a data set of alpha grain crystal orientations (e.g., the measured three Euler angles) originate from the same prior beta grain. For example, determining whether alpha phase data points originate from the same prior beta grain based on whether the orientations of the beta phase determined from the measured alpha phase data points match or are misoriented relative to each other. Determining whether certain data points of a measured alpha phase data set may be used to determine missing or inaccurate measurements of the alpha phase, e.g., to determine missing or inaccurate third Euler angles, as described further below. In examples described herein, a total amount of misorientation of the beta phase, e.g., integrated across a plurality of data points of a measured data set, may be used as a metric, or merit function, of a method or algorithm for determining missing and/or inaccurate third Euler angles of data sets measured by measurement device **14**.

[0073] FIG. **5A** is a plot of an example total beta phase misorientation curve **500** as a function of iteration number of an iterative algorithm for determining third Euler angles from a data set of

measured orientations of a material. In the example shown, at iteration number **1**, the total beta phase misorientation is high due to missing and/or inaccurate third Euler angles of the data set. For example, a user may cause measurement device **14** to measure a data set including an image of the crystal orientations of sample **22**, the orientations being characterized by the three Euler angles, and where sample **22** may be an alloy such as Ti-6Al-4V. Data analysis device **12** may receive the data set and transform at least a portion of the data points to the beta phase, via Burgers orientation relationship **210**, a total misorientation of the beta phase after transformation. Data analysis device **12** may then perform an iterative method (described further below) to refine the missing and/or inaccurate third Euler angles of the data set and transform at least a portion of the refined data set to the beta phase and determine the total misorientation of the beta phase. Data analysis device **12** may continue to perform the method until a stop condition is reached. The stop condition may be after execution of the method for a predetermined number of iterations, after the total beta phase misorientation is less than or equal to a threshold value, or any other suitable stop condition.

[0074] In the example shown, data analysis device **12** may iterate until iteration number **504**. Beta phase misorientation curve **500** may have a minimum beta phase misorientation value **502** at iteration number **506**. In some examples, if measurement device **14** measures a portion of sample **16** originating from a single prior beta grain, the minimum value **502** approaches zero. In other examples, if measurement device **14** measures a portion of sample **16** originating from a plurality of prior beta grains, the minimum value **502** approaches a minimum asymptote value due to the prior beta grains having different orientations. For example, as shown in FIG. 5B, measurement device **14** may measure a data set of alpha phase orientations within area **650** of sample **16**, e.g., area **650** including a plurality of alpha grains **612**, **614**, **616**, **622**, **624**, **626**, **628**, **632**, and **634** originating from prior beta grains **600**, **602**, and **604**, and other alpha grains within area **650** originating from other prior beta grains (not shown). Data analysis device **12** may be configured to determine which data points of the data set correspond to which alpha grains **612-634** originated from which prior beta grains **600**, **602**, and **604** based at least in part on determining a minimum total beta phase misorientation value **502**.

[0075] FIG. 6A is a schematic diagram illustrating an example measurement area **650**. For example, measurement device **14** may be configured to measure a plurality pixels (e.g., alpha data set **652**) corresponding to a plurality of x-y positions of measurement area **650** corresponding to an “image” of the crystal orientations of sample **16**. Each pixel of the plurality of pixels corresponding to the image of area **650** may include a vector of information, for example, each of three Euler angles, an x-position, and a y-position. The plurality of pixels, and information contained within each pixel, may alternatively be referred to as alpha phase data set **652** or alpha phase image **652**.

[0076] Measurement device **14** may be configured to output alpha phase data set **652** to data analysis device **12** or **27**, or alternatively measurement device **14** may be configured to output one or more signals indicative of the measurements corresponding to alpha phase data set **652** data analysis device **12** or **27** may form alpha phase data set **652** based on the one or more signals. Additionally, data analysis device **12**, analysis device **12**, or control module **28** may add information to data set **652**, e.g., data analysis device **12** or **27** may determine the presence of, and which pixels are associated with, one or more alpha phase grains based on the x-y positions and Euler angles of the plurality of pixels, and may associate each pixel of the plurality of pixels with the one or more alpha grains. For example, data analysis device **12** or **31** may add an alpha grain identification (ID) value to the vector of information of each pixel.

[0077] In some examples, data analysis device **12** or **31** may determine which alpha phase grains may have originated from the same prior beta phase grain based on the x-y positions, Euler angles, and alpha grain IDs of the plurality of pixels, and may associate each pixel of the plurality of pixels with a prior beta grain. For example, data analysis device **12** or **31** may add a prior beta grain identification (ID) value to the vector of information of each pixel, and each pixel of data set **652** may include a vector of information including each of three Euler angles, an x-position, a y-

position, an alpha grain ID, and a prior beta grain ID.

[0078] FIGS. 7-16 are flow diagrams of an example technique or techniques of determining third Euler angles (ϕ_2) of a measured data set including erroneous or missing third Euler angles, and determining the presence and size statistics of MTRs, and are described together below. FIG. 7 is a flow diagram of an example method which data analysis device 12 (or data analysis device 31) may perform to determine the third Euler angles of a data set measured by measurement device 14, which may in turn be used to determine the presence and size statistics of MTRs. FIG. 7 is described with concurrent reference to system 10 of FIG. 1 and system 27 of FIG. 2, although other systems may be used.

[0079] In some examples, the techniques or techniques of FIGS. 7-16 include determining, by control circuitry (e.g., analysis module 32), alpha phase grains and alpha phase grain boundaries of the measurement of sample 16 based on the x-y positions and first and second Euler angles (ϕ_1 , Φ), determining which alpha phases are from the same prior beta grains, and adjusting the third Euler angles (ϕ_2) of the measurement based on reducing a total misorientation of the crystals of the prior beta grains.

[0080] Control module 28 may receive a data set 652 including a plurality of measured Euler angles at each of a plurality positions of sample 16 (700). For example, a user 42 may select a measurement option via input device 40 causing control module 28 to control measurement device 14 to measure sample 16 and output the measurement as a data set 652 including a plurality of measured Euler angles as a function of spatial position of a portion of the area of surface 22. In some examples, the alpha phase data set 652 may include a plurality of pixels arranged in a grid or an array as shown in FIG. 6. Each pixel of alpha phase data set 652 may include a vector of information including an x-position, a y-position, a first Euler angle (ϕ_1), a second Euler angle (Φ), a third Euler angle (ϕ_2) which may be erroneous or missing, an alpha grain ID which may initially be unknown, and a prior beta grain ID which may also initially be unknown. Control module 28 may invoke analysis module 32 to process the received alpha phase data set 652.

[0081] Analysis module 32 may zero the measured third Euler angle (ϕ_2) values and determine the alpha grain IDs of each of the plurality of pixels (702). In some examples, measurement device 14 may be a polarized light measurement device or a SRAS measurement device and may output alpha phase data set 652 with third Euler angle (ϕ_2) values already set to zero, and analysis device 32 may not need to zero the measured third Euler angle (ϕ_2) values. In other examples, measurement device 14 may output alpha data set 652 without any measured third Euler angle (ϕ_2) values, e.g., missing values, or with partially missing measured third Euler angle (ϕ_2) values, with erroneous measured third Euler angle (ϕ_2) values, or with partially missing and/or partially erroneous measured third Euler angle (ϕ_2) values.

[0082] For example, analysis module 32 may replace the measured, and erroneous or missing, third Euler angle (ϕ_2) value of each pixel with a zero value. Analysis module 32 may determine the alpha grain IDs of each of the plurality of pixels based on the x-y positions and first and second Euler angles (ϕ_1 , Φ) of the plurality of alpha phase data set 652 and replace and/or add the alpha grain ID value determined for each pixel. This work may be performed with an algorithm developed in MATLAB®, free-ware packages like DREAM3D, or bespoke codes that compare individual pixels and their relative misorientation to develop unique alpha phase grain ID's that correlate clusters of pixels to an ID number.

[0083] Analysis module 32 may determine which alpha grains may be from the same prior beta grain (704). Analysis module 32 may determine which pixels of alpha phase data set 652 may be from the same prior beta grain based on the first and second Euler angles (ϕ_1 , Φ) and the alpha grain IDs of each pixel of the plurality of pixels. For example, analysis module 32 may compare two adjacent (e.g., touching) pixels from different alpha grains and having different alpha grain ID values, such as pixels 662 and 664 of FIG. 6A.

[0084] Method step (704) is described below with reference to FIGS. 8, 6A, 6B, and 17. FIG. 8 is a

flow diagram of an example method which data analysis device **12** (or data analysis device **31**) may perform to determine which alpha phase grains may be from the same prior beta grain. FIG. **6B** is a schematic diagram of an example database module **34** configured to store a plurality of values, data sets, data, and/or information. FIG. **17** illustrates an example rotation matrix $g(\phi_1, \Phi, \phi_2)$ which may be used to determine the crystal axes (x' , y' , z') and specimen directions (X , Y , Z) of a crystal relative to the specimen coordinate system.

[0085] Referring to FIG. **8**, analysis module **32** may randomly select a first pixel of a first alpha grain (**802**). For example, analysis module **32** may randomly select pixel **662** of alpha grain **614**. In some examples, at (**802**), analysis module **32** may initiate (e.g., replace with an initial value such as “1” or “0”) an iteration counter **770** (FIG. **6B**). For example, analysis module **32** may perform one or more steps of the methods described herein iteratively and may exit an iterative loop of one or more steps based on an exit condition being satisfied. In some examples, the exit condition may be a value, or count or iteration number, of iteration counter **770**.

[0086] Analysis module **32** may then determine if there is a second pixel adjacent to (e.g., touching) the first pixel **662** that is from a different alpha grain, e.g., has a different alpha grain ID associated with that adjacent pixel (**804**). If analysis module **32** determines that there is not a second pixel adjacent to the first pixel and from a different alpha grain (the NO branch at **804**), analysis module **32** may randomly select a different first pixel of the same or a different alpha grain at (**802**), e.g., the method may loop back to (**802**). If analysis module **32** determines that there is a second pixel adjacent to the first pixel and from a different alpha grain (the YES branch at **804**), analysis module **32** may determine if the second pixel is from the same prior beta grain as the first pixel (**806**).

[0087] For example, at (**804**), analysis module **32** may determine that second pixel **664** is adjacent to first pixel **662** and is from a different alpha grain **616** than first pixel **662**. At (**806**), analysis module **32** may determine whether pixel **662**, of alpha grain **616**, and pixel **664**, of alpha grain **662**, may be from (e.g., associated with) the same prior beta grain based on a difference in the orientation (also referred to as a misorientation) of the c-axes of the crystals represented by pixels **662**, **664**. For example, the three Euler angles of each of the first and second pixels pixel **662**, **664** may be relative to a reference, or specimen, coordinate system which may be defined by the orientation of sample **16** and measurement device **14** during the measurement of sample **16**, and which may serve as a global reference coordinate frame for the pixels of alpha phase data set **652**. The Euler angles may be used to determine the orientation of certain axes of symmetry and the planes of faces of the crystalline structure of sample **16** relative to the specimen coordinate system, e.g., via a rotation matrix $g(\phi_1, \Phi, \phi_2)$.

[0088] FIG. **17** illustrates an example rotation matrix $g(\phi_1, \Phi, \phi_2)$ which may be used to determine the crystal axes (x' , y' , z') and specimen directions (X , Y , Z) of a crystal relative to the specimen coordinate system. The c-axis of a hcp crystal, e.g., of the alpha phase of sample **16**, corresponds to the z' crystal axes. Analysis module **32** may determine the misorientation between the c-axes of pixels by calculating the dot product between the z' crystal axes of each pixel, e.g., the dot product of the third row (e.g., elements a_{31} , a_{32} , a_{33}) of $g(\phi_1, \Phi, \phi_2)$ of each pixel. In the example shown in FIG. **17**, the z' axis, or c-axis, of each pixel of alpha phase data set **652** may be calculated based on only the first two Euler angles, e.g., (ϕ_1, Φ) , and does not depend on the erroneous or missing third Euler angle (ϕ_2) .

[0089] As described above, a bcc prior beta grain **100** (FIG. **3**) of a titanium alloy such as Ti-6Al-4V may transform to only twelve orientations of and hcp alpha phase. Of the twelve orientations, there are only three different c-axis misorientation angles allowed, namely, 0° , 60° , and 90° . For example, two different alpha grains having phase transformed from the same prior beta grain may have c-axis orientations that differ only by 0° , 60° , or 90° . Analysis module **32** may determine whether pixel **662** and pixel **664** may be from the same prior beta grain based on the c-axis misorientation between pixel **662** and pixel **664** being substantially equal to 0° , 30° , or 90° , e.g., to

within the measurement error of measurement device **14**.

[0090] For example, if analysis module **32** determines that pixel **668** is the second pixel, analysis module **32** may determine that pixel **662** and pixel **664** are not from the same prior beta grain **600** (the NO branch at **806**) based on the c-axis misorientation between pixels **662**, **664** not being any one of 0°, 60°, and 90° plus or minus 0.5°, or plus or minus 1°, or plus or minus 2.5°, or plus or minus 5°, or plus or minus any suitable measurement error. Analysis module **32** may then randomly select a different first pixel of the same or a different alpha grain at (**802**), e.g., the method may loop back to (**802**).

[0091] Alternatively, analysis module **32** may determine that pixel **662** and pixel **664** are from the same prior beta grain **600** (the YES branch at **806**) based on the c-axis misorientation between pixels **662**, **664** being any one of 0°, 60°, and 90° plus or minus 0.5°, or plus or minus 10, or plus or minus 2.5°, or plus or minus 5°, or plus or minus any suitable measurement error. For example, analysis module **32** may determine that pixel **662** and pixel **664** may be from the same prior beta grain **600**. The method may then advance to (**706**) FIG. 7, e.g., via advancing to (**902**) of FIG. 9, as described below.

[0092] Analysis module **32** may adjust the value of the third Euler angle (ϕ_2) and determine a Burgers inverse transformation solution that reduces a total beta phase misorientation (**706**). For example, analysis module **32** may assign a value for third Euler angle (ϕ_2) for a pair of pixels, e.g., pixels **662**, **664**, that are from different alpha phase grains **64**, **616** and that may be from the same prior beta grain, e.g., prior beta grain **600**, based on the c-axis misorientation between the pixels **662**, **664**.

[0093] Analysis module **32** may then select an inverse transformation (IT) solution, e.g., IT operator, that satisfies the Burgers orientation relationship for the alpha and beta phases of a material, e.g., referred to herein as a Burgers IT solution. For example, analysis module **32** may adjust the value of the third Euler angles (ϕ_2) of all of the pixels of alpha grain **614** to the same value as pixel **662** and adjust the value of the third Euler angles (ϕ_2) of all of the pixels of alpha grain **616** to the same value as pixel **664**. Analysis module **32** may then inverse transform all of the pixels of alpha grain **614** via the selected first Burgers IT solution to a beta phase, and analysis module **32** may inverse transform all of the pixels of alpha grain **616** via the selected second Burgers IT solution to the beta phase.

[0094] The plurality of transformed pixels may be a portion of the plurality of pixels of a beta phase data set **752** (FIG. 6B) corresponding to beta phase of the measurement area **650** sample **16** that underwent a phase transformation to the alpha phase that was measured to yield alpha phase data set **652**. The plurality of pixels of beta phase data set **752** may correspond, in x-y positions, to the x-y positions of the plurality of pixels of measured alpha phase data set **652**. Each pixel of the plurality of pixels of beta phase data set **752** may include an x-position and a y-position, each of which may be the same as the x, y position of its corresponding pixel of the measure alpha phase data set **652**, three beta phase Euler angles indicating the orientation of the body-centered cubic beta phase crystal represented by a pixel of the beta phase data set **752**. Each pixel of the plurality of pixels of beta phase data set **752** may also include an alpha grain ID value (which may be the same as the alpha grain ID value of its corresponding pixel of the measure alpha phase data set **652**), and a beta grain ID value (which may be the same as the beta grain ID value of its corresponding pixel of the measure alpha phase data set **652**). In some examples, beta phase data set **752** may be a plurality of data sets, one for each iteration, e.g., beta phase data set **752** may be saved as a stack of images as a function of iteration number corresponding to the current total beta phase misorientation **780** and Burgers IT solutions dataset **762** of the same iteration number.

[0095] After inverse transforming the pixels of the first and second alpha grains **614**, **616**, analysis module **32** may determine a current total beta phase misorientation **780** (FIG. 6B) of the beta phase, e.g., based on the beta phase Euler angles of each of the pixels of the beta phase data set. If the adjusted values of the third Euler angles (ϕ_2) of the pixels of alpha grains **614** and **616** are

correct, the pixels from the different alpha grains **614** and **616** will inverse transform to the same prior beta grain, e.g., a single crystal having the same orientation in the areas corresponding to both alpha grains **614**, **616**. If the adjusted values of the third Euler angles (ϕ_2) of the pixels of alpha grains **614** and **616** are incorrect but closer to being correct, the pixels from the different alpha grains **614** and **616** will inverse transform to the same prior beta grain having orientations that are more similar in the areas corresponding to both alpha grains **614**, **616**, and the total beta phase misorientation of beta phase data set **752** will decrease.

[0096] Method step (**706**) is further described below with reference to FIGS. **9** and **10**. FIG. **9** is a flow diagram of method step (**706**), e.g., an example method of adjusting the value of the third Euler angle (ϕ_2), determining a beta phase data set, and determining a Burgers inverse transformation solution that reduces a total beta phase misorientation. FIG. **10** is a schematic diagram illustrating the six unique Burgers IT solutions.

[0097] Referring to FIG. **9**, analysis module **32** may determine the c-axis misorientation of the first pixel and the second pixel (**902**). For example, analysis module **32** may determine the c-axis misorientation between first pixel **662** and second pixel **664** to be about 0° , 60° , and 90° , e.g., as described above at (**806**). If analysis module **32** determines the c-axis misorientation between first pixel **662** and second pixel **664** to be about 0° , analysis module **32** may assign a random angle value to the third Euler angle (ϕ_2) of pixel **662** and the same random angle value to the third Euler angle (ϕ_2) of pixel **664** (**904**). If analysis module **32** determines the c-axis misorientation between first pixel **662** and second pixel **664** to be about 60° , analysis module **32** may assign a random angle value to the third Euler angle (ϕ_2) of pixel **662** and a different angle value to the third Euler angle (ϕ_2) of pixel **664** such that the x' direction (FIG. **17**) direction is 600 different between the pixels **662**, **664**, e.g., to maintain the 60° c.-axis misorientation after having changed the value of the third Euler angle (ϕ_2) of pixel **662** (**906**). If analysis module **32** determines the c-axis misorientation between first pixel **662** and second pixel **664** to be about 90° , analysis module **32** may assign a random angle value to the third Euler angle (ϕ_2) of pixel **662** and a different angle value to the third Euler angle ($\phi_{sup.2}$) of pixel **664** such that the x' direction (FIG. **17**) direction is 90° different between the pixels **662**, **664**, e.g., to maintain the 90° c.-axis misorientation after having changed the value of the third Euler angle (ϕ_2) of pixel **662** (**908**). For each of (**904**)-(**908**), analysis module **32** may assign the random angle value to the third Euler angle (ϕ_2) of the first pixel, e.g., pixel **662**, that is from 0° to 360° . For each of (**904**)-(**908**), analysis module **32** may assign the different angle value to the third Euler angle (ϕ_2) of the second pixel, e.g., pixel **664**, that is that is cyclic from 0° to 360° . For example, if analysis module **32** assigns a different angle value of the third Euler angle (ϕ_2) of pixel **664** that is less than 0° or greater than 360° , analysis module **32** may add or subtract a multiple of 360° to the different angle value is such that the different angle value is equal to or greater than 0° and less than or equal to 360° .

[0098] Analysis module **32** may then assign the random angle value of the third Euler angle (ϕ_2) of pixel **662** to all of the pixels of the first alpha grain having the same alpha grain ID as pixel **662**, e.g., alpha grain **614** (FIG. **6A**), and analysis module **32** may assign the angle value of the third Euler angle (ϕ_2) of pixel **664** to all of the pixels of the second alpha grain having the same alpha grain ID as pixel **664**, e.g., alpha grain **616**.

[0099] Analysis module **32** may then inverse transform all of the first alpha grain pixels and all of the second alpha grain pixels to the beta phase according to each of nine combinations of three of the Burgers IT solutions (**912**). For example, analysis module **32** may inverse transform all of the first alpha grain pixels and all of the second alpha grain pixels to the beta phase according to a subset of less than all six of a set of six unique Burgers IT solutions, e.g., the first three Burgers IT solutions **1002-1006** (FIG. **10**). For a single alpha phase orientation of an hcp crystal, there are six unique ways to transform the alpha phase data (e.g., orientation) into a beta phase crystal (e.g., of the prior beta grain). FIG. **10** illustrates a graphical representation of the six Burgers IT solutions **1002-1012**. Each of **1002-1012** is a two-dimensional (2D) representation of a hcp crystal overlaid

with different orientations of a resulting bcc crystal after having inverse transformed the hcp crystal according to the Burgers IT solution associated with each 2D representation. The six unique Burgers IT solutions may be six unique matrices that may be applied to a vector of the three Euler angles characterizing the orientation of the alpha phase hcp crystal in order to calculate one of the six possible orientations of the bcc beta phase crystal from which the alpha phase crystal transformed. The calculation results in a vector of three beta phase Euler angles characterizing one of the six possible orientations of the bcc beta phase crystal. For example, a known (or measured) crystal orientation of the alpha phase of a hcp crystal, such as Ti-6Al-4V, may have originated from one of six beta phase crystal orientations of a bcc prior beta grain, and the Burgers IT solutions may be used to inverse transform the known (or measured) alpha phase orientation to one of the six possible beta phase orientations from which the alpha phase crystal transformed.

[0100] Analysis module **32** may inverse transform the pixels of the first alpha grain, e.g., alpha grain **614**, by applying one of the first three Burgers IT solutions **1002-1006** to the Euler angles of each pixel of alpha grain **614**. Analysis module **32** may inverse transform the pixels of the second alpha grain, e.g., alpha grain **616**, by applying one of the first three Burgers IT solutions **1002-1006** (e.g., the same or a different one as applied to alpha grain **612**) to the Euler angles of each pixel of alpha grain **616**. Analysis module **32** may may inverse transform the pixels of the first and second alpha grains **614**, **616** using all nine combinations (e.g., variants **1-9** of the table shown in FIG. **10**) of the first three Burgers IT solutions **1002-1006**, resulting in nine potential beta phase data sets corresponding to the nine variant combinations of Burgers IT solutions **1002-1006** illustrated in the last column of the table of FIG. **10**.

[0101] Analysis module **32** may determine which variant combination **1-9** of the first three Burgers IT solutions **1002-1006** results in the lowest total beta phase misorientation (**914**). For example, analysis module **32** may compare the orientations of all of corresponding pixels of the nine potential beta phase data sets resulting from each inverse transformation from all nine variant combinations and sum (e.g., integrate) all of the misorientations of all of the pixels for each of the nine beta phase data sets.

[0102] In some examples, analysis module **32** may then select the combination of Burgers IT solutions associated with the beta phase data set having the lowest total beta phase misorientation. Analysis module **32** may then associate the lowest total beta phase misorientation with the current value of the iteration counter **770**, index the iteration counter (e.g., add “1” to the iteration value), and associate the selected Burgers IT solutions combination (e.g., resulting in the lowest total beta phase misorientation) of the current iteration with the respective alpha grains **614**, **616** (**916**). For example, analysis module **32** may cause current total beta phase misorientation **780** (FIG. **6B**) to be updated with the value of the lowest total beta phase misorientation of the current iteration. In some examples, analysis module **32** may save a plurality of current total beta phase misorientations **780**, e.g., as a function of iteration number. For example, current total beta phase misorientation **780** may be an array of saved lowest total beta phase misorientation numbers, each associated with an iteration number. Analysis module **32** may also save and/or update beta phase dataset **752** to include the beta phase data set corresponding to the lowest total beta phase misorientation of the current iteration. Analysis module **32** may save and/or update Burgers IT solutions data set **762** with the Burgers IT solutions corresponding to the lowest total beta phase misorientation of the current iteration.

[0103] For example, analysis module **32** may determine that variant **4** (FIG. **10**) results in the lowest total beta phase misorientation, e.g., applying Burgers IT solution **1002** to the pixels of alpha grain **614** and applying the Burgers IT solution **1004** to the pixels of alpha grain **616** results in the lowest lowest total beta phase misorientation of the nine variants. Analysis module **32** may then index iteration counter **770** in database **34**. Analysis module **32** may save and/or update beta phase data set **752** to be the beta phase data set corresponding to the lowest total beta phase misorientation of the current iteration. Analysis module **32** may save and/or update at least a portion

of Burgers IT solutions data set **762** with Burgers IT solutions **1002, 1004**. For example, Burgers IT solutions data set **762** may include a plurality of pixels corresponding to the pixels of alpha phase data set **652**, e.g., each pixel including the same x-y positions, alpha grain ID, and beta phase grain ID as each corresponding pixel of alpha phase data set **652**. Each pixel of Burgers IT solutions data set **762** may also include a Burgers IT solution and/or solution identifier, e.g., one of Burgers IT solutions **1002-1012**. For example, analysis module **32** may update only the pixels of Burgers IT solutions data set **762** corresponding to the alpha grain ID of the first alpha phase pixel, e.g., alpha phase pixel **662**, with Burgers IT solution **1002**, analysis module **32** may update only the pixels of Burgers IT solutions data set **762** corresponding to the alpha grain ID of the second alpha phase pixel, e.g., alpha phase pixel **664**, with Burgers IT solution **1004**.

[0104] In some examples, Burgers IT solutions data set **762** may be a plurality of data sets, one for each iteration, e.g., Burgers IT solutions data set **762** may be saved as a stack of images as a function of iteration number corresponding to the current total beta phase misorientation **780** and beta phase misorientation data set **752** of the same iteration number. For example, if the last iteration of the method of FIGS. **7-16** does not result in the lowest total beta phase misorientation across all iterations, the lowest beta phase misorientation may be saved and associated with the iteration number of the lowest beta phase misorientation, such that the beta phase misorientation data set **752** and Burgers IT solutions data set **762** of the same iteration number may be recovered.

[0105] In some examples, the lowest total beta phase misorientation of the current iteration may not be lower than the lowest total beta phase misorientation of a previous iteration. For example, the method of FIGS. **7-16** may allow the total beta phase misorientation to increase as a function of iteration number (or value), e.g., so as to be able to “exit” a saddle point (e.g., local minimum), and progress to a solution (e.g., a set of third Euler angles of data set **652** and Burgers IT solutions data set **752**) that is lower than the local minimum. Analysis module **32** may execute the method of FIGS. **7-16** to allow the total beta phase misorientation to increase as a function of iteration count value, at least for a number of iterations, in order to exit a saddle point or local minimum and progress to a global or lower local minimum of the total beta phase misorientation.

[0106] Analysis module **32** may determine if an exit condition is met (**918**). In some examples, analysis module **32** may determine that the exit condition is met when the value of iteration counter **770** is equal to an iteration exit count, e.g., a threshold number of iterations. For example, analysis module **32** may determine an iteration exit count based on the number of alpha grains of data set **652**. In some examples, analysis module **32** may determine the iteration exit count to be the number of alpha grains of data set **652**, or some multiple or fraction of the number of alpha grains of data set **652**. In other examples, analysis module **32** may determine that the exit condition is met when the lowest or current total beta phase misorientation **780** is less than or equal to a beta phase misorientation threshold value.

[0107] If analysis module **32** determines that the exit condition is not met (the NO branch at (**918**), analysis module **32** may randomly select a different first pixel of the same or a different alpha grain at (**802**), e.g., the method may loop back to (**802**). For example, method steps (**704**) and (**706**), which may comprise method steps (**802**) through (**918**), may be performed in an iterative loop in which both the next pixel of the next iteration is selected randomly, and values to replace for the third Euler angle of the selected pixel (and pixels of the same alpha grain) are selected randomly as a Monte Carlo type numerical search method. If analysis module **32** determines that the exit condition is met (the YES branch at **918**), the method may then advance to (**708**) of FIG. **7**, e.g., via advancing to (**1102**) of FIG. **11**, as described below.

[0108] FIGS. **11-13** are flow diagrams of example method steps of performing method step (**708**) of FIG. **7**. The method of each of FIGS. **11-13** are methods of refining the selection of the Burgers IT solutions that reduce the total beta phase misorientation for a given alpha phase data set **652** with adjusted or perturbed third Euler angle values. The methods of FIGS. **11-13** do not adjust or perturb the third Euler angle values, but rather describe the steps of refining the selection of a

combination Burgers IT solutions to be applied to the identified alpha grains of the alpha phase data set **652**, e.g., before further perturbing and refining the values of the third Euler angles of the identified alpha grains. In some examples, the methods of FIGS. **11-13** may be performed as separate iterative loops. In the examples shown, FIGS. **11-13** may iterate through at least all of the alpha grains of alpha data set **652**, e.g., as opposed to performing a Monte Carlo type numerical search method in which pixels of alpha grains are randomly selected until an exit condition is met. In some examples, even though FIGS. **11-13** may be performed as separate iterative loops (e.g., from each other as well as from FIGS. **8-9**), analysis module **32** may continue to track a global iteration count, e.g., iteration counter **770**, across all iteration loops of the method of FIGS. **7-16**. [0109] FIG. **11** is a flow diagram of an example method of refining selection of the first three Burgers IT solutions **1002-1006**. FIG. **11** is described with concurrent reference to system **10** of FIGS. **1**, although other systems, such as system **27** illustrated in FIG. **2**, may be used.

[0110] Analysis module **32** may select an alpha grain and a first pixel of the selected alpha grain (**1102**). For example, analysis module **32** may select first pixel **662** of selected alpha grain **614** (FIG. **6A**). Analysis module **32** may select a second pixel of a second alpha grain different from the selected alpha grain, and the second pixel also being adjacent to (e.g., “touching”) the first pixel, and where the second pixel also may be from the same prior beta grain as the first pixel (**1104**). For example, analysis module **32** may select second pixel **664** of second alpha grain **616**, and both selected alpha grain **614** and different alpha grain **616** may be from prior beta grain **600** (FIG. **6A**).

[0111] Analysis module **32** may inverse transform all of the pixels of the same alpha grain as the first pixel according to all nine combinations of the first three Burgers IT solutions (**1106**). For example, analysis module **32** may transform the updated alpha phase data set **652** to nine variants of a beta phase data set as described above at method step (**912**). Analysis module **32** may then determine whether the total beta phase misorientation of any of the nine variants of the beta phase data set reduces the current total beta phase misorientation **780** (**1108**). For example, analysis module **32** may determine the total beta phase misorientation of each of the nine variants of beta phase data sets, determine which variant is the lowest, and compare the lowest total beta phase misorientation of the nine variants to the current total beta phase misorientation value **780** stored in database module **34** at (**916**) or at (**1110**) of a previous iteration.

[0112] If analysis module **32** determines that the lowest total beta phase misorientation of the nine variants of beta phase data sets is lower than the current total beta phase misorientation value **780** (e.g., does reduce the total beta phase misorientation, the YES branch at **1108**), analysis module **32** may save the lowest total beta phase misorientation of the nine variants to the current total beta phase misorientations **780**, index the iteration counter (e.g., add “1” to the iteration value), and associate the Burgers IT solutions combination resulting in the lowest total beta phase misorientation alpha grains **614**, **616** (**1110**). Analysis module **32** may also save and/or update beta phase dataset **752** to include the beta phase data set corresponding to the lowest total beta phase misorientation of the nine variants. Analysis module **32** may save and/or update Burgers IT solutions data set **762** with the Burgers IT solutions corresponding to the lowest total beta phase misorientation of the nine variants. The method may then advance to step (**1112**) described below.

[0113] If analysis module **32** determines that the lowest total beta phase misorientation of the nine variants of beta phase data sets is not lower than the current total beta phase misorientation value **780** (e.g., does not reduce the total beta phase misorientation, the NO branch at **1108**), analysis module **32** may determine whether there is another pixel of alpha phase data set **652** that is adjacent to the first pixel, e.g., pixel **662**, whether that pixel is from a different alpha grain than the first pixel, e.g., has a different alpha grain ID than pixel **662**, and whether that pixel may be from the same prior beta grain, e.g., has the same prior beta grain ID as pixel **662** (**1112**). In some examples, the method may advance to step (**1112**) from the NO branch at (**1108**) without indexing iteration counter **770**.

[0114] If analysis module **32** determines that there is another pixel of alpha phase data set **652** that

is adjacent to the first pixel and from a different alpha grain (the YES branch at **1112**), analysis module **32** may then select that pixel as the second pixel (**1114**), and the method may loop back to (**1106**). If analysis module **32** determines that there is not another pixel of alpha phase data set **652** that is adjacent to the first pixel and from a different alpha grain and the same prior beta grain (the NO branch at **1112**), analysis module **32** may then determine whether there is another pixel of the selected alpha grain that is different from the first pixel that was not previously selected as a first pixel (**1116**). For example, analysis module **32** may store the x-y position of pixel **662** in database **34** as having been selected, and may determine whether there are other pixels having the same alpha grain ID as pixel **662** that have not been selected.

[0115] If analysis module **32** determines that there is another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the YES branch at **1116**), analysis module **32** may select that pixel as a first pixel (**1118**) and the method may loop back to (**1104**). If analysis module **32** determines that there is not another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the NO branch at **1116**), analysis module **32** may then determine whether there is another alpha grain that was not previously selected (**1120**). For example, analysis module **32** may store the alpha grain ID of alpha grain **614** in database **34** as having been selected, and may determine whether there are other alpha grain IDs that have not, e.g., alpha grain **616**.

[0116] If analysis module **32** determines that there is another alpha grain that was not previously selected (the YES branch at **1120**), analysis module **32** may select that alpha grain select a first of the newly selected alpha grain (**1102**), e.g., the method may loop back to (**1102**). If analysis module **32** determines that there is not another alpha grain that was not previously selected (the NO branch at **1120**), the method may advance to (**1202**).

[0117] In the example shown, analysis module **32** may execute method steps (**1102**) through (**1120**) to refine the selection of the Burgers IT solutions and refine Burgers IT solutions data set **762** using just the first three Burgers IT solutions **1002**, **1004**, **1006**, e.g., before perturbing values of the third Euler angles of alpha phase data set **652** to search for the correct third Euler angles. As described below, analysis module **32** may execute method steps (**1202**) through (**1220**) to refine the selection of the Burgers IT solutions, and refine Burgers IT solutions data set **762**, using just the second three Burgers IT solutions **1008**, **1010**, **1012**, and analysis module **32** may execute method steps (**1302**) through (**1320**) to refine the selection of the Burgers IT solutions, and refine Burgers IT solutions data set **762**, using all six Burgers IT solutions **1002-1012**.

[0118] FIG. **12** is a flow diagram of an example method of refining selection of the second three Burgers IT solutions **1008-1012**. FIG. **12** is described with concurrent reference to system **10** of FIGS. **1**, although other systems, such as system **27** illustrated in FIG. **2**, may be used.

[0119] Analysis module **32** may select an alpha grain and a first pixel of the selected alpha grain (**1202**). For example, analysis module **32** may select first pixel **662** of selected alpha grain **614** (FIG. **6A**). Analysis module **32** may select a second pixel of a second alpha grain different from the selected alpha grain, and the second pixel also being adjacent to (e.g., “touching”) the first pixel, and where the second pixel also may be from the same prior beta grain as the first pixel (**1204**). For example, analysis module **32** may select second pixel **664** of second alpha grain **616**, and both selected alpha grain **614** and different alpha grain **616** may be from prior beta grain **600** (FIG. **6A**).

[0120] Analysis module **32** may inverse transform all of the pixels of the same alpha grain as the first pixel according to all nine combinations of the second three Burgers IT solutions (**1206**). For example, analysis module **32** may inverse transform all of the same alpha grain as the first pixel according to a subset of less than all six of a set of six unique Burgers IT solutions, e.g., the second three Burgers IT solutions **1008-1012** (FIG. **10**). In some examples, analysis module **32** may transform the updated alpha phase data set **652** to nine variants of a beta phase data set as described above at method step (**912**), except replacing the first three Burgers IT solutions with the second three Burgers IT solutions **1008-1012**. Analysis module **32** may then determine whether the total

beta phase misorientation of any of the nine variants of the beta phase data set reduces the current total beta phase misorientation **780 (1208)**. For example, analysis module **32** may determine the total beta phase misorientation of each of the nine variants of beta phase data sets, determine which variant is the lowest, and compare the lowest total beta phase misorientation of the nine variants to the current total beta phase misorientation value **780** stored in database module **34** at **(916)** or at **(1210)** of a previous iteration.

[0121] If analysis module **32** determines that the lowest total beta phase misorientation of the nine variants of beta phase data sets is lower than the current total beta phase misorientation value **780** (e.g., does reduce the total beta phase misorientation, the YES branch at **1208**), analysis module **32** may save the lowest total beta phase misorientation of the nine variants to the current total beta phase misorientations **780**, index the iteration counter (e.g., add “1” to the iteration value), and associate the Burgers IT solutions combination resulting in the lowest total beta phase misorientation alpha grains **614, 616 (1210)**. Analysis module **32** may also save and/or update beta phase dataset **752** to include the beta phase data set corresponding to the lowest total beta phase misorientation of the nine variants. Analysis module **32** may save and/or update Burgers IT solutions data set **762** with the Burgers IT solutions corresponding to the lowest total beta phase misorientation of the nine variants. The method may then advance to step **(1212)** described below.

[0122] If analysis module **32** determines that the lowest total beta phase misorientation of the nine variants of beta phase data sets is not lower than the current total beta phase misorientation value **780** (e.g., does not reduce the total beta phase misorientation, the NO branch at **1208**), analysis module **32** may determine whether there is another pixel of alpha phase data set **652** that is adjacent to the first pixel, e.g., pixel **662**, whether that pixel is from a different alpha grain than the first pixel, e.g., has a different alpha grain ID than pixel **662**, and whether that pixel may be from the same prior beta grain, e.g., has the same prior beta grain ID as pixel **662 (1212)**. In some examples, the method may advance to step **(1212)** from the NO branch at **(1208)** without indexing iteration counter **770**.

[0123] If analysis module **32** determines that there is another pixel of alpha phase data set **652** that is adjacent to the first pixel and from a different alpha grain (the YES branch at **1212**), analysis module **32** may then select that pixel as the second pixel **(1214)**, and the method may loop back to **(1206)**. If analysis module **32** determines that there is not another pixel of alpha phase data set **652** that is adjacent to the first pixel and from a different alpha grain and the same prior beta grain (the NO branch at **1212**), analysis module **32** may then determine whether there is another pixel of the selected alpha grain that is different from the first pixel that was not previously selected as a first pixel **(1216)**. For example, analysis module **32** may store the x-y position of pixel **662** in database **34** as having been selected, and may determine whether there are other pixels having the same alpha grain ID as pixel **662** that have not been selected.

[0124] If analysis module **32** determines that there is another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the YES branch at **1216**), analysis module **32** may select that pixel as a first pixel **(1218)** and the method may loop back to **(1204)**. If analysis module **32** determines that there is not another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the NO branch at **1216**), analysis module **32** may then determine whether there is another alpha grain that was not previously selected **(1220)**. For example, analysis module **32** may store the alpha grain ID of alpha grain **614** in database **34** as having been selected, and may determine whether there are other alpha grain IDs that have not, e.g., alpha grain **616**.

[0125] If analysis module **32** determines that there is another alpha grain that was not previously selected (the YES branch at **1220**), analysis module **32** may select that alpha grain select a first of the newly selected alpha grain **(1202)**, e.g., the method may loop back to **(1202)**. If analysis module **32** determines that there is not another alpha grain that was not previously selected (the NO branch at **1220**), the method may advance to **(1302)**.

[0126] FIG. 13 is a flow diagram of an example method of refining selection of all six of Burgers IT solutions **1002-1006**. FIG. 13 is described with concurrent reference to system **10** of FIGS. 1, although other systems, such as system **27** illustrated in FIG. 2, may be used.

[0127] Analysis module **32** may select an alpha grain and a first pixel of the selected alpha grain (**1302**). For example, analysis module **32** may select first pixel **662** of selected alpha grain **614** (FIG. 6A). Analysis module **32** may select a second pixel of a second alpha grain different from the selected alpha grain, and the second pixel also being adjacent to (e.g., “touching”) the first pixel, and where the second pixel also may be from the same prior beta grain as the first pixel (**1304**). For example, analysis module **32** may select second pixel **664** of second alpha grain **616**, and both selected alpha grain **614** and different alpha grain **616** may be from prior beta grain **600** (FIG. 6A).

[0128] Analysis module **32** may inverse transform all of the pixels of the same alpha grain as the first pixel according to 36 combinations of all six Burgers IT solutions (**1306**). For example, analysis module **32** may transform the updated alpha phase data set **652** to 36 variants of a beta phase data set, similar to as described above at method step (**912**), except using all six of the Burgers IT solutions **1002-1012**. Analysis module **32** may then determine whether the total beta phase misorientation of any of the 36 variants of the beta phase data set reduces the current total beta phase misorientation **780** (**1308**). For example, analysis module **32** may determine the total beta phase misorientation of each of the 36 variants of beta phase data sets, determine which variant is the lowest, and compare the lowest total beta phase misorientation of the 36 variants to the current total beta phase misorientation value **780** stored in database module **34** at (**916**) or at (**1310**) of a previous iteration.

[0129] If analysis module **32** determines that the lowest total beta phase misorientation of the 36 variants of beta phase data sets is lower than the current total beta phase misorientation value **780** (e.g., does reduce the total beta phase misorientation, the YES branch at **1308**), analysis module **32** may save the lowest total beta phase misorientation of the 36 variants to the current total beta phase misorientations **780**, index the iteration counter (e.g., add “1” to the iteration value), and associate the Burgers IT solutions combination resulting in the lowest total beta phase misorientation alpha grains **614**, **616** (**1310**). Analysis module **32** may also save and/or update beta phase dataset **752** to include the beta phase data set corresponding to the lowest total beta phase misorientation of the 36 variants. Analysis module **32** may save and/or update Burgers IT solutions data set **762** with the Burgers IT solutions corresponding to the lowest total beta phase misorientation of the 36 variants. The method may then advance to step (**1312**) described below.

[0130] If analysis module **32** determines that the lowest total beta phase misorientation of the 36 variants of beta phase data sets is not lower than the current total beta phase misorientation value **780** (e.g., does not reduce the total beta phase misorientation, the NO branch at **1308**), analysis module **32** may determine whether there is another pixel of alpha phase data set **652** that is adjacent to the first pixel, e.g., pixel **662**, whether that pixel is from a different alpha grain than the first pixel, e.g., has a different alpha grain ID than pixel **662**, and whether that pixel may be from the same prior beta grain, e.g., has the same prior beta grain ID as pixel **662** (**1312**). In some examples, the method may advance to step (**1312**) from the NO branch at (**1308**) without indexing iteration counter **770**.

[0131] If analysis module **32** determines that there is another pixel of alpha phase data set **652** that is adjacent to the first pixel and from a different alpha grain (the YES branch at **1312**), analysis module **32** may then select that pixel as the second pixel (**1314**), and the method may loop back to (**1306**). If analysis module **32** determines that there is not another pixel of alpha phase data set **652** that is adjacent to the first pixel and from a different alpha grain and the same prior beta grain (the NO branch at **1312**), analysis module **32** may then determine whether there is another pixel of the selected alpha grain that is different from the first pixel that was not previously selected as a first pixel (**1316**). For example, analysis module **32** may store the x-y position of pixel **662** in database **34** as having been selected, and may determine whether there are other pixels having the same

alpha grain ID as pixel **662** that have not been selected.

[0132] If analysis module **32** determines that there is another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the YES branch at **1316**), analysis module **32** may select that pixel as a first pixel (**1318**) and the method may loop back to (**1304**). If analysis module **32** determines that there is not another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the NO branch at **1316**), analysis module **32** may then determine whether there is another alpha grain that was not previously selected (**1320**). For example, analysis module **32** may store the alpha grain ID of alpha grain **614** in database **34** as having been selected, and may determine whether there are other alpha grain IDs that have not, e.g., alpha grain **616**.

[0133] If analysis module **32** determines that there is another alpha grain that was not previously selected (the YES branch at **1320**), analysis module **32** may select that alpha grain select a first of the newly selected alpha grain (**1302**), e.g., the method may loop back to (**1302**). If analysis module **32** determines that there is not another alpha grain that was not previously selected (the NO branch at **1320**), the method may advance to (**710**).

[0134] FIGS. **14-16** are flow diagrams of example method steps of performing method step (**710**) of FIG. **7**. The method of each of FIGS. **14-16** are methods of refining the values of the third Euler angle values of alpha phase data set **652** by iteratively perturbing the values by different amounts, and then further refining selection of the Burgers IT solutions after each refinement/perturbation of a third Euler angle. In some examples, the methods of FIGS. **14-16** may be performed as separate iterative loops. In the examples shown, FIGS. **14-16** may iterate through at least all of the alpha grains of alpha data set **652**, e.g., as opposed to performing a Monte Carlo type numerical search method in which pixels of alpha grains are randomly selected until an exit condition is met. In some examples, even though FIGS. **14-16** may be performed as separate iterative loops (e.g., from each other as well as from FIGS. **8-9**), analysis module **32** may continue to track a global iteration count, e.g., iteration counter **770**, across all iteration loops of the method of FIGS. **7-16**.

[0135] FIG. **14** is a flow diagram of an example method of refining values of third Euler angles as well as selection of all six Burgers IT solutions **1002-1006**. FIG. **14** is described with concurrent reference to system **10** of FIGS. **1**, although other systems, such as system **27** illustrated in FIG. **2**, may be used.

[0136] Analysis module **32** may select an alpha grain and a first pixel of the selected alpha grain (**1402**). For example, analysis module **32** may select first pixel **662** of selected alpha grain **614** (FIG. **6A**). Analysis module **32** may select a second pixel of a second alpha grain different from the selected alpha grain, and the second pixel also being adjacent to (e.g., “touching”) the first pixel, and where the second pixel also may be from the same prior beta grain as the first pixel (**1404**). For example, analysis module **32** may select second pixel **664** of second alpha grain **616**, and both selected alpha grain **614** and different alpha grain **616** may be from prior beta grain **600** (FIG. **6A**).

[0137] Analysis module **32** may perturb (e.g., change or adjust) the value of the third Euler angle (ϕ_2) of the first pixel by a first perturbation amount (**1405**). For example, analysis module **32** may add or subtract a perturbation amount of 30 degrees to the third Euler angle (ϕ_2) of pixel **662**. In some examples, control module **32** perturbs the value of the third Euler angles (ϕ_2) of all of the pixels of the same alpha grain ID is the first pixel, e.g., all of the pixels of alpha grain **614**.

[0138] Analysis module **32** may inverse transform all of the pixels of the same alpha grain as the first pixel according to 36 combinations of all six Burgers IT solutions (**1406**). For example, analysis module **32** may transform the updated alpha phase data set **652** to 36 variants of a beta phase data set, similar to as described above at method step (**912**), except using all six of the Burgers IT solutions **1002-1012**. Analysis module **32** may then determine whether the total beta phase misorientation of any of the 36 variants of the beta phase data set reduces the current total beta phase misorientation **780** (**1408**). For example, analysis module **32** may determine the total beta phase misorientation of each of the 36 variants of beta phase data sets, determine which

variant is the lowest, and compare the lowest total beta phase misorientation of the 36 variants to the current total beta phase misorientation value **780** stored in database module **34** at (**916**) or at (**1410**) of a previous iteration.

[0139] If analysis module **32** determines that the lowest total beta phase misorientation of the 36 variants of beta phase data sets is lower than the current total beta phase misorientation value **780** (e.g., does reduce the total beta phase misorientation, the YES branch at **1408**), analysis module **32** may save the lowest total beta phase misorientation of the 36 variants to the current total beta phase misorientations **780**, index the iteration counter (e.g., add “1” to the iteration value), and associate the Burgers IT solutions combination resulting in the lowest total beta phase misorientation alpha grains **614**, **616** (**1410**). Analysis module **32** may also save and/or update beta phase dataset **752** to include the beta phase data set corresponding to the lowest total beta phase misorientation of the 36 variants. Analysis module **32** may save and/or update Burgers IT solutions data set **762** with the Burgers IT solutions corresponding to the lowest total beta phase misorientation of the 36 variants. The method may then advance to step (**1412**) described below. If analysis module **32** determines that the lowest total beta phase misorientation of the nine variants of beta phase data sets is not lower than the current total beta phase misorientation value **780** (e.g., does not reduce the total beta phase misorientation, the NO branch at **1408**), analysis module **32** may determine whether there is another pixel of alpha phase data set **652** that is adjacent to the first pixel, e.g., pixel **662**, whether that pixel is from a different alpha grain than the first pixel, e.g., has a different alpha grain ID than pixel **662**, and whether that pixel may be from the same prior beta grain, e.g., has the same prior beta grain ID as pixel **662** (**1412**). In some examples, the method may advance to step (**1412**) from the NO branch at (**1408**) without indexing iteration counter **770**.

[0140] If analysis module **32** determines that there is another pixel of alpha phase data set **652** that is adjacent to the first pixel and from a different alpha grain (the YES branch at **1412**), analysis module **32** may then select that pixel as the second pixel (**1414**), and the method may loop back to (**1406**). If analysis module **32** determines that there is not another pixel of alpha phase data set **652** that is adjacent to the first pixel and from a different alpha grain and the same prior beta grain (the NO branch at **1412**), analysis module **32** may then determine whether there is another pixel of the selected alpha grain that is different from the first pixel that was not previously selected as a first pixel (**1416**). For example, analysis module **32** may store the x-y position of pixel **662** in database **34** as having been selected, and may determine whether there are other pixels having the same alpha grain ID as pixel **662** that have not been selected.

[0141] If analysis module **32** determines that there is another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the YES branch at **1416**), analysis module **32** may select that pixel as a first pixel (**1418**) and the method may loop back to (**1404**). If analysis module **32** determines that there is not another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the NO branch at **1416**), analysis module **32** may then determine whether there is another alpha grain that was not previously selected (**1420**). For example, analysis module **32** may store the alpha grain ID of alpha grain **614** in database **34** as having been selected, and may determine whether there are other alpha grain IDs that have not, e.g., alpha grain **616**.

[0142] If analysis module **32** determines that there is another alpha grain that was not previously selected (the YES branch at **1420**), analysis module **32** may select that alpha grain select a first of the newly selected alpha grain (**1402**), e.g., the method may loop back to (**1402**). If analysis module **32** determines that there is not another alpha grain that was not previously selected (the NO branch at **1420**), the method may advance to (**1502**).

[0143] FIG. **15** is a flow diagram of an example method of further refining values of third Euler angles as well as selection of all six Burgers IT solutions **1002-1006**. FIG. **15** is described with concurrent reference to system **10** of FIGS. **1**, although other systems, such as system **27** illustrated in FIG. **2**, may be used.

[0144] Analysis module 32 may select an alpha grain and a first pixel of the selected alpha grain (1502). For example, analysis module 32 may select first pixel 662 of selected alpha grain 614 (FIG. 6A). Analysis module 32 may select a second pixel of a second alpha grain different from the selected alpha grain, and the second pixel also being adjacent to (e.g., “touching”) the first pixel, and where the second pixel also may be from the same prior beta grain as the first pixel (1504). For example, analysis module 32 may select second pixel 664 of second alpha grain 616, and both selected alpha grain 614 and different alpha grain 616 may be from prior beta grain 600 (FIG. 6A). [0145] Analysis module 32 may perturb (e.g., change or adjust) the value of the third Euler angle (ϕ_2) of the first pixel by a second perturbation amount (1505). For example, analysis module 32 may add or subtract a perturbation amount of about 10.5 degrees to the third Euler angle (ϕ_2) of pixel 662. In some examples, control module 32 perturbs the value of the third Euler angles (ϕ_2) of all of the pixels of the same alpha grain ID is the first pixel, e.g., all of the pixels of alpha grain 614.

[0146] Analysis module 32 may inverse transform all of the pixels of the same alpha grain as the first pixel according to all nine combinations of the first three Burgers IT solutions (1506). For example, analysis module 32 may transform the updated alpha phase data set 652 to nine variants of a beta phase data set as described above at method step (912). Analysis module 32 may then determine whether the total beta phase misorientation of any of the nine variants of the beta phase data set reduces the current total beta phase misorientation 780 (1508). For example, analysis module 32 may determine the total beta phase misorientation of each of the nine variants of beta phase data sets, determine which variant is the lowest, and compare the lowest total beta phase misorientation of the nine variants to the current total beta phase misorientation value 780 stored in database module 34 at (916) or at (1510) of a previous iteration.

[0147] If analysis module 32 determines that the lowest total beta phase misorientation of the nine variants of beta phase data sets is lower than the current total beta phase misorientation value 780 (e.g., does reduce the total beta phase misorientation, the YES branch at 1508), analysis module 32 may save the lowest total beta phase misorientation of the nine variants to the current total beta phase misorientations 780, index the iteration counter (e.g., add “1” to the iteration value), and associate the Burgers IT solutions combination resulting in the lowest total beta phase misorientation alpha grains 614, 616 (1510). Analysis module 32 may also save and/or update beta phase dataset 752 to include the beta phase data set corresponding to the lowest total beta phase misorientation of the nine variants. Analysis module 32 may save and/or update Burgers IT solutions data set 762 with the Burgers IT solutions corresponding to the lowest total beta phase misorientation of the nine variants. The method may then advance to step (1512) described below.

[0148] If analysis module 32 determines that the lowest total beta phase misorientation of the nine variants of beta phase data sets is not lower than the current total beta phase misorientation value 780 (e.g., does not reduce the total beta phase misorientation, the NO branch at 1508), analysis module 32 may determine whether there is another pixel of alpha phase data set 652 that is adjacent to the first pixel, e.g., pixel 662, whether that pixel is from a different alpha grain than the first pixel, e.g., has a different alpha grain ID than pixel 662, and whether that pixel may be from the same prior beta grain, e.g., has the same prior beta grain ID as pixel 662 (1512). In some examples, the method may advance to step (1512) from the NO branch at (1508) without indexing iteration counter 770.

[0149] If analysis module 32 determines that there is another pixel of alpha phase data set 652 that is adjacent to the first pixel and from a different alpha grain (the YES branch at 1512), analysis module 32 may then select that pixel as the second pixel (1514), and the method may loop back to (1506). If analysis module 32 determines that there is not another pixel of alpha phase data set 652 that is adjacent to the first pixel and from a different alpha grain and the same prior beta grain (the NO branch at 1512), analysis module 32 may then determine whether there is another pixel of the selected alpha grain that is different from the first pixel that was not previously selected as a first

pixel (**1516**). For example, analysis module **32** may store the x-y position of pixel **662** in database **34** as having been selected, and may determine whether there are other pixels having the same alpha grain ID as pixel **662** that have not been selected.

[0150] If analysis module **32** determines that there is another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the YES branch at **1516**), analysis module **32** may select that pixel as a first pixel (**1518**) and the method may loop back to (**1504**). If analysis module **32** determines that there is not another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the NO branch at **1516**), analysis module **32** may then determine whether there is another alpha grain that was not previously selected (**1520**). For example, analysis module **32** may store the alpha grain ID of alpha grain **614** in database **34** as having been selected, and may determine whether there are other alpha grain IDs that have not, e.g., alpha grain **616**.

[0151] If analysis module **32** determines that there is another alpha grain that was not previously selected (the YES branch at **1520**), analysis module **32** may select that alpha grain select a first of the newly selected alpha grain (**1502**), e.g., the method may loop back to (**1502**). If analysis module **32** determines that there is not another alpha grain that was not previously selected (the NO branch at **1520**), the method may advance to (**1602**).

[0152] FIG. **16** is a flow diagram of an example method of further refining values of third Euler angles as well as selection of all six Burgers IT solutions **1002-1006**. FIG. **16** is described with concurrent reference to system **10** of FIGS. **1**, although other systems, such as system **27** illustrated in FIG. **2**, may be used.

[0153] Analysis module **32** may select an alpha grain and a first pixel of the selected alpha grain (**1602**). For example, analysis module **32** may select first pixel **662** of selected alpha grain **614** (FIG. **6A**). Analysis module **32** may select a second pixel of a second alpha grain different from the selected alpha grain, and the second pixel also being adjacent to (e.g., “touching”) the first pixel, and where the second pixel also may be from the same prior beta grain as the first pixel (**1604**). For example, analysis module **32** may select second pixel **664** of second alpha grain **616**, and both selected alpha grain **614** and different alpha grain **616** may be from prior beta grain **600** (FIG. **6A**).

[0154] Analysis module **32** may perturb (e.g., change or adjust) the value of the third Euler angle (ϕ_2) of the first pixel by a third perturbation amount (**1605**). For example, analysis module **32** may add or subtract a perturbation amount of about 5.2 degrees to the third Euler angle (ϕ_2) of pixel **662**. In some examples, control module **32** perturbs the value of the third Euler angles (ϕ_2) of all of the pixels of the same alpha grain ID is the first pixel, e.g., all of the pixels of alpha grain **614**.

[0155] Analysis module **32** may inverse transform all of the pixels of the same alpha grain as the first pixel according to all nine combinations of the first three Burgers IT solutions (**1606**). For example, analysis module **32** may transform the updated alpha phase data set **652** to nine variants of a beta phase data set as described above at method step (**912**). Analysis module **32** may then determine whether the total beta phase misorientation of any of the nine variants of the beta phase data set reduces the current total beta phase misorientation **780** (**1608**). For example, analysis module **32** may determine the total beta phase misorientation of each of the nine variants of beta phase data sets, determine which variant is the lowest, and compare the lowest total beta phase misorientation of the nine variants to the current total beta phase misorientation value **780** stored in database module **34** at (**916**) or at (**1610**) of a previous iteration.

[0156] If analysis module **32** determines that the lowest total beta phase misorientation of the nine variants of beta phase data sets is lower than the current total beta phase misorientation value **780** (e.g., does reduce the total beta phase misorientation, the YES branch at **1608**), analysis module **32** may save the lowest total beta phase misorientation of the nine variants to the current total beta phase misorientations **780**, index the iteration counter (e.g., add “1” to the iteration value), and associate the Burgers IT solutions combination resulting in the lowest total beta phase misorientation alpha grains **614**, **616** (**1610**). Analysis module **32** may also save and/or update beta

phase dataset **752** to include the beta phase data set corresponding to the lowest total beta phase misorientation of the nine variants. Analysis module **32** may save and/or update Burgers IT solutions data set **762** with the Burgers IT solutions corresponding to the lowest total beta phase misorientation of the nine variants. The method may then advance to step **(1612)** described below. [0157] If analysis module **32** determines that the lowest total beta phase misorientation of the nine variants of beta phase data sets is not lower than the current total beta phase misorientation value **780** (e.g., does not reduce the total beta phase misorientation, the NO branch at **1608**), analysis module **32** may determine whether there is another pixel of alpha phase data set **652** that is adjacent to the first pixel, e.g., pixel **662**, whether that pixel is from a different alpha grain than the first pixel, e.g., has a different alpha grain ID than pixel **662**, and whether that pixel may be from the same prior beta grain, e.g., has the same prior beta grain ID as pixel **662** (**1612**). In some examples, the method may advance to step **(1612)** from the NO branch at **(1608)** without indexing iteration counter **770**.

[0158] If analysis module **32** determines that there is another pixel of alpha phase data set **652** that is adjacent to the first pixel and from a different alpha grain (the YES branch at **1612**), analysis module **32** may then select that pixel as the second pixel (**1614**), and the method may loop back to **(1606)**. If analysis module **32** determines that there is not another pixel of alpha phase data set **652** that is adjacent to the first pixel and from a different alpha grain and the same prior beta grain (the NO branch at **1612**), analysis module **32** may then determine whether there is another pixel of the selected alpha grain that is different from the first pixel that was not previously selected as a first pixel (**1616**). For example, analysis module **32** may store the x-y position of pixel **662** in database **34** as having been selected, and may determine whether there are other pixels having the same alpha grain ID as pixel **662** that have not been selected.

[0159] If analysis module **32** determines that there is another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the YES branch at **1616**), analysis module **32** may select that pixel as a first pixel (**1618**) and the method may loop back to **(1604)**. If analysis module **32** determines that there is not another pixel of the selected alpha grain that is different from the first pixel and that was not previously selected as a first pixel (the NO branch at **1616**), analysis module **32** may then determine whether there is another alpha grain that was not previously selected (**1620**). For example, analysis module **32** may store the alpha grain ID of alpha grain **614** in database **34** as having been selected, and may determine whether there are other alpha grain IDs that have not, e.g., alpha grain **616**.

[0160] If analysis module **32** determines that there is another alpha grain that was not previously selected (the YES branch at **1620**), analysis module **32** may select that alpha grain select a first of the newly selected alpha grain (**1602**), e.g., the method may loop back to **(1602)**. If analysis module **32** determines that there is not another alpha grain that was not previously selected (the NO branch at **1620**), the method may advance to **(712)**.

[0161] Analysis module **32** may determine the third Euler angles (ϕ_2) of alpha phase data set **652** (**712**). For example, analysis module **32** may determine the third Euler angles (ϕ_2) by retrieving the refined third Euler angles (ϕ_2) values of each of the pixels from database **34**. In other examples, analysis module **32** may determine the third Euler angles (ϕ_2) based on the adjusted/refined beta phase data set **752** and the adjusted/refined Burgers IT solution data set **762**. Analysis module **32** may determine the presence and/or size statistics of one or more MTR based on alpha phase data set **652** with corrected third Euler angles (**714**). In some examples, analysis module **32** may output an alert based on determining the MTR, e.g., to output device **38**.

[0162] In some examples, a blade, or other component, of a gas turbine engine includes sample **16**, and analysis module **32** may determine a status of the blade based on at least one of alpha phase data set **652** or the MTR and output the status of the blade, e.g., to output device **38**. For example, analysis module **32** may determine a PASS or FAIL status of the fan blade based on the presence and/or size statistics of one or more MTR, e.g., the status being indicative of the robustness of the

material of sample **16** and the blade and resistance to failure, such as a cold dwell failure.

[0163] FIG. **18** is a set of two-dimensional heat maps illustrating the results of the x' , y' , and z' axes of an alpha phase data set including third Euler angles and corresponding beta phase data set determined via the method of FIG. **7** compared with the results of the x' , y' , and z' axes of an alpha phase data set and corresponding beta phase data set measured via electron backscatter diffraction, each for an area including mostly the same single prior beta grain. FIG. **19** is a table of the misorientations and axis direction for alpha phase orientations from a single prior beta grain corresponding to the alpha phase data set of FIG. **18**. In the example shown in FIG. **18**, the deduced (e.g., determined via the method of FIG. **7**) beta phase (e.g., the determined prior beta grain) was deduced from an EBSD measured alpha phase data with the third Euler angle removed, and is shown to be correct based on implication of the Burgers relationship in terms of pole figure correspondence of the $(110)_{\text{sub.}\beta}$ and the $(0001)_{\text{sub.}\alpha}$ pole figures (FIG. **20**). FIG. **20** is an illustration of the correspondence of the deduced $(110)_{\text{sub.}\beta}$ pole figure with the $(0001)_{\text{sub.}\alpha}$ third Euler angle deduced and measured pole figures. The deduced beta phase is shown to be correct also based on implication of the Burgers relationship in terms of pole figure correspondence of the $(111)_{\text{sub.}\beta}$ and the $(11-20)_{\text{sub.}\alpha}$ pole figures (FIG. **21**). FIG. **21** is an illustration of the correspondence of the deduced $(111)_{\text{sub.}\beta}$ pole figure with the $(11-20)_{\text{sub.}\alpha}$ third Euler angle deduced and measured pole figures.

[0164] FIG. **22** is a set of two-dimensional heat maps illustrating the results of the x' , y' , and z' axes of an alpha phase data set including third Euler angles and corresponding beta phase data set determined via the method of FIG. **7** compared with the results of the x' , y' , and z' axes of an alpha phase data set and corresponding beta phase data set measured via electron backscatter diffraction, each for an area including the same plurality of prior beta grains.

[0165] In some examples, because the systems, devices, and methods disclosed herein generate all pole figures, prior beta grain orientations, and inverse pole figure maps correctly, the methodology may be used to measured MTR distributions accurately and may be used in conjunction with techniques such as polarized light to characterize the complete crystallographic orientation specimens optically.

[0166] The devices, systems, and methods disclosed herein may be used to calculate the underlying beta grain orientation, e.g., corresponding to a measured alpha phase data set, with an understanding of several concepts. For example, for an arbitrary rotation matrix $g(\phi_1, \Phi, \phi_2)$, the direction of the c-axis or the $[001]$ crystal direction (e.g., z' direction) in the rotated reference form may be defined by the bottom row (e.g., third row) of the rotation matrix (FIG. **17**), and the dot product of the bottom row of two arbitrary rotation matrices may define the misorientation of the c-axis of two hcp crystals. Once a third Euler angle ($\phi_{\text{sup.}2}$) is defined, a product of the top row of two arbitrary rotation matrices may define the mis-orientation of the x' directions for two arbitrary crystals. Once a third Euler angle ($\phi_{\text{sup.}2}$) is defined, a product of the middle row of two arbitrary rotation matrices $g(\phi_1, \Phi, \phi_2)$ may define the misorientation of the y' directions for two arbitrary crystals.

[0167] In some examples, the interplanar angles of (110) planes in the beta phase are restricted to 0, 60, and 90 degrees, the interplanar angles of (111) planes in the beta phase are 0 and 70 degrees, 32 seconds, the interplanar angles between the (110) and (111) planes in the beta phase are 35 degrees, 16 seconds and 90 degrees, and the $(11-20)$ and $(10-10)$ planes of the hcp phase are 30 degrees apart from one another.

[0168] In some examples, to align an alpha hcp crystal to a beta crystal that obeys the Burgers relation where the c-axis of the hcp crystals are 0° or 180° misoriented, the third Euler angles (ϕ_2) must be set equal to one another, e.g., because of the horizontal and vertical diads for the $(110)_{\text{sub.}\beta}$ and $(111)_{\text{sub.}\beta}$ poles. FIG. **23** is an illustration of plots of the alpha and beta phase orientations for two adjacent pixels of alpha data set **652** having a c-axis misorientation of 0° or 180° . In some examples, for c-axes of two adjacent pixels of alpha data set **652** that are 90° degrees

misoriented, their third Euler angles (ϕ_2) are 30° apart in the hcp alpha phase to align x' direction. FIG. 24 is an illustration of plots of the alpha and beta phase orientations for two adjacent pixels of alpha data set 652 having a c-axis misorientation of 30° . In some examples, for c-axes of two adjacent pixels of alpha data set 652 that are 60° degrees misoriented, their third Euler angles (ϕ_2) are 60° apart in the hcp alpha phase system because of the horizontal cubic mirror plane for (111).sub. β and (110).sub. β poles. FIG. 25 is an illustration of plots of the alpha and beta phase orientations for two adjacent pixels of alpha data set 652 having a c-axis misorientation of 60° . In some examples, because of the cubic and hexagonal symmetries, when it comes to x' and y' in a rotation matrix, any third Euler angle (ϕ_2) orientations could be 30° off (which is the equivalent of 90° in the hcp alpha phase due to its symmetry). FIG. 26 is an illustration of plots of possible 30° third Euler angle (ϕ_2) misorientations due to cubic and hexagonal symmetries in relation to a cartesian coordinate system.

[0169] In some examples, the method of FIG. 7 may use symmetry operators. FIG. 27 is a table of cubic symmetry operators, and FIG. 28 is a table of hexagonal symmetry operators.

[0170] The following numbered examples may illustrate one or more aspects of this disclosure:

[0171] Example 1: A method includes: measuring, by at least one of a polarized light device, a spatially resolved acoustic spectroscopy device, or an eddy current device, an alpha phase data set indicative of an alpha phase of a crystalline structure of a material; receiving, by processing circuitry, the alpha phase data set, wherein the alpha phase data set comprises a plurality of pixels, wherein each pixel of the plurality of pixels includes a position, a first Euler angle (ϕ_1), a second Euler angle (Φ), and a third Euler angle (ϕ_2), wherein the third Euler angle (ϕ_2) is missing or erroneous; adjusting, by the processing circuitry, the third Euler angle (ϕ_2) of a pixel of the plurality of pixels; and storing, by the processing circuitry and based on adjusting the third Euler angle (ϕ_2) of the pixel reducing a total beta phase misorientation, the alpha phase data set.

[0172] Example 2: The method of example 1, wherein the material comprises a titanium alloy.

[0173] Example 3: The method of example 1 or example 2, wherein the pixel is a first pixel, the method further includes determining, by the processing circuitry, an alpha grain of each pixel of the plurality of pixels based on a position of each pixel and the first and second Euler angles (ϕ_1 , Φ) of each pixel; determining, by the processing circuitry, which alpha grains of the plurality of pixels may be from the same prior beta grain, and adjusting, by the processing circuitry, the third Euler angle (ϕ_2) of a second pixel of the plurality of pixels, wherein adjusting the third Euler angle (ϕ_2) of the first pixel and the second pixel is based on: the first and second pixels are from different alpha grains, the first and second pixels being from the same prior beta grain, and the first and second pixels being adjacent to each other.

[0174] Example 4: The method of example 3, wherein determining which alpha grains of the plurality of pixels may be from the same prior beta grain comprises: randomly selecting a pixel of the plurality of pixels as the first pixel, wherein the first pixel is from a first alpha grain; selecting the second pixel based on the second pixel being from a second alpha grain different from the first alpha grain and the second pixel being adjacent to the first pixel; and determining that a c-axis misorientation between the first and second pixels is one of about 0° , about 60° , or about 90° .

[0175] Example 5: The method of example 4, further includes assigning, by the processing circuitry, a random angle value to the third Euler angle (ϕ_2) of the first pixel; assigning, by the processing circuitry, a different angle value to the third Euler angle (ϕ_2) of the second pixel such that the c-axis misorientation between the first and second pixels is one of about 0° , about 60° , or about 90° ; assigning, by the processing circuitry, the random angle value to the third Euler angle (ϕ_2) of each pixel of the plurality of pixels that is from the first alpha grain; and assigning, by the processing circuitry, the different angle value to the third Euler angle (ϕ_2) of each pixel of the plurality of pixels that is from the second alpha.

[0176] Example 6: The method of example 5, further includes determining, by the processing circuitry, a first Burgers inverse transform (IT) solution to transform the first pixel to a first beta

phase pixel of a beta phase data set and a second Burgers inverse transform (IT) solution to transform the second pixel to a second beta phase pixel of the beta phase data set, wherein determining the first and second Burgers IT solutions is based on reducing the total beta phase misorientation of the beta phase data set after transforming the first and second pixels by the first and second Burgers IT solutions.

[0177] Example 7: The method of example 6, wherein the first and second Burgers IT solutions are selected from a subset of less than all six of a set of six unique Burgers IT solutions that transform an alpha phase crystal to a beta phase crystal.

[0178] Example 8: The method of example 7, further includes determining, by the processing circuitry, a Burgers IT solution from the subset of less than all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution; and subsequently determining, by the processing circuitry, a Burgers IT solution from the subset including all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution.

[0179] Example 9: The method of example 8, further includes changing, by the processing circuitry, the value of the third Euler angle (ϕ_2) of the first pixel by at least one of 30° , 10.53° , or 5.16° ; and determining, by the processing circuitry, a Burgers IT solution from the subset including all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution.

[0180] Example 10: The method of example 9, wherein a blade of a gas turbine engine comprises the material, the method further includes determining, by the processing circuitry and subsequent to changing the value of the third Euler angle (ϕ_2) of the first pixel, at least one microtexture region (MTR) based on the alpha phase data set; outputting, by the processing circuitry, an alert based on determining the at least one MTR; determining, by the processing circuitry, a status of the blade based on at least one of the alpha phase data set or the at least one MTR; and outputting, by the processing circuitry, the status of the blade.

[0181] Example 11: A device includes: a measurement device configured to measure a crystalline structure of a material and output an alpha phase data set indicative of the measurement of the crystalline structure; a memory; and one or more processors implemented in circuitry and in communication with the memory, the one or more processors configured to: receive the alpha phase data set, wherein the alpha phase data set comprises a plurality of pixels, wherein each pixel of the plurality of pixels includes a position, a first Euler angle (ϕ_1), a second Euler angle (Φ), and a third Euler angle (ϕ_2), wherein the third Euler angle (ϕ_2) is missing or erroneous; adjust the third Euler angle (ϕ_2) of a pixel of the plurality of pixels; and store, based on adjusting the third Euler angle (ϕ_2) of the pixel reducing a total beta phase misorientation, the alpha phase data set.

[0182] Example 12: The device of example 11, wherein the material comprises a titanium alloy.

[0183] Example 13: The device of example 11 or example 12, wherein the pixel is a first pixel, wherein the one or more processors are further configured to: determine an alpha grain of each pixel of the plurality of pixels based on a position of each pixel and the first and second Euler angles (ϕ_1 , Φ) of each pixel; determine which alpha grains of the plurality of pixels may be from the same prior beta grain, and adjust the third Euler angle (ϕ_2) of a second pixel of the plurality of pixels, wherein adjusting the third Euler angle (ϕ_2) of the first pixel and the second pixel is based on: the first and second pixels are from different alpha grains, the first and second pixels being from the same prior beta grain, and the first and second pixels being adjacent to each other.

[0184] Example 14: The device of example 13, wherein, to determine which alpha grains of the plurality of pixels may be from the same prior beta grain, the one or more processors are configured to: randomly select a pixel of the plurality of pixels as the first pixel, wherein the first

pixel is from a first alpha grain; select the second pixel based on the second pixel being from a second alpha grain different from the first alpha grain and the second pixel being adjacent to the first pixel; and determine that a c-axis misorientation between the first and second pixels is one of about 00, about 600, or about 900.

[0185] Example 15: The device of example 14, wherein the one or more processors are further configured to: assign a random angle value to the third Euler angle (ϕ_2) of the first pixel; assign a different angle value to the third Euler angle (ϕ_2) of the second pixel such that the c-axis misorientation between the first and second pixels is one of about 00, about 600, or about 900; assign the random angle value to the third Euler angle (ϕ_2) of each pixel of the plurality of pixels that is from the first alpha grain; and assign the different angle value to the third Euler angle (ϕ_2) of each pixel of the plurality of pixels that is from the second alpha.

[0186] Example 16: The device of example 15, wherein the one or more processors are further configured to: determine a first Burgers inverse transform (IT) solution to transform the first pixel to a first beta phase pixel of a beta phase data set and a second Burgers inverse transform (IT) solution to transform the second pixel to a second beta phase pixel of the beta phase data set, wherein determining the first and second Burgers IT solutions is based on reducing the total beta phase misorientation of the beta phase data set after transforming the first and second pixels by the first and second Burgers IT solutions.

[0187] Example 17: The device of example 16, wherein the first and second Burgers IT solutions are selected from a subset of less than all six of a set of six unique Burgers IT solutions that transform an alpha phase crystal to a beta phase crystal.

[0188] Example 18: The device of example 17, wherein the one or more processors are further configured to: determine a Burgers IT solution from the subset of less than all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution; and subsequently determine a Burgers IT solution from the subset including all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution.

[0189] Example 19: The device of example 18, wherein a blade of a gas turbine engine comprises the material, wherein the one or more processors are further configured to: change the value of the third Euler angle (ϕ_2) of the first pixel by at least one of 30°, 10.53°, or 5.16°; determine a Burgers IT solution from the subset including all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution; determine, subsequent to changing the value of the third Euler angle (ϕ_2) of the first pixel, at least one microtexture region (MTR) based on the alpha phase data set; output and alert based on determining the at least one MTR; determine a status of the blade based on at least one of the alpha phase data set or the at least one MTR; and output the status of the blade.

[0190] Example 20: A non-transitory computer-readable storage medium having stored thereon instructions that, when executed, configure a processor to: receive an alpha phase data set, wherein the alpha phase data set comprises a plurality of pixels, wherein each pixel of the plurality of pixels includes a position, a first Euler angle (ϕ_1), a second Euler angle (Φ), and a third Euler angle (ϕ_2), wherein the third Euler angle (ϕ_2) is missing or erroneous; adjust the third Euler angle (ϕ_2) of a pixel of the plurality of pixels; and store, based on adjusting the third Euler angle (ϕ_2) of the pixel reducing a total beta phase misorientation, the alpha phase data set.

[0191] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware or any combination thereof. For example, various aspects of the described techniques may be implemented within processing circuitry, which may include one or more processors, including one or more microprocessors, digital signal processors (DSPs),

application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also form one or more processors or processing circuitry configured to perform one or more of the techniques of this disclosure.

[0192] Such hardware, software, and firmware may be implemented, and various operation may be performed within the same device, within separate devices, and/or on a coordinated basis within, among or across several devices, to support the various operations and functions described in this disclosure. In addition, any of the described units, circuits or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as circuits or units is intended to highlight different functional aspects and does not necessarily imply that such circuits or units must be realized by separate hardware or software components. Rather, functionality associated with one or more circuits or units may be performed by separate hardware or software components or integrated within common or separate hardware or software components. Processing circuitry described in this disclosure, including a processor or multiple processors, may be implemented, in various examples, as fixed-function circuits, programmable circuits, or a combination thereof. Fixed-function circuits refer to circuits that provide particular functionality with preset operations. Programmable circuits refer to circuits that can be programmed to perform various tasks and provide flexible functionality in the operations that can be performed. For instance, programmable circuits may execute software or firmware that cause the programmable circuits to operate in the manner defined by instructions of the software or firmware. Fixed-function circuits may execute software instructions (e.g., to receive stimulation parameters or output stimulation parameters), but the types of operations that the fixed-function circuits perform are generally immutable. In some examples, one or more of the units may be distinct circuit blocks (fixed-function or programmable), and in some examples, one or more of the units may be integrated circuits.

[0193] The techniques described in this disclosure may also be embodied or encoded in a computer-readable medium, such as a computer-readable storage medium, containing instructions that may be described as non-transitory media. Instructions embedded or encoded in a computer-readable storage medium may cause a programmable processor, or other processor, to perform the method, e.g., when the instructions are executed. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a CD-ROM, a floppy disk, a cassette, magnetic media, optical media, or other computer readable media.

[0194] Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. A method comprising: measuring, by at least one of a polarized light device, a spatially resolved acoustic spectroscopy device, or an eddy current device, an alpha phase data set indicative of an alpha phase of a crystalline structure of a material; receiving, by processing circuitry, the alpha phase data set, wherein the alpha phase data set comprises a plurality of pixels, wherein each pixel of the plurality of pixels includes a position, a first Euler angle (ϕ_1), a second Euler angle Φ , and a third Euler angle (ϕ_2), wherein the third Euler angle (ϕ_2) is missing or erroneous; adjusting, by the processing circuitry, the third Euler angle (ϕ_2) of a pixel of the plurality of pixels; and storing, by the processing circuitry and based on adjusting the third Euler angle (ϕ_2) of the pixel reducing a total beta phase misorientation, the alpha phase data set.

2. The method of claim 1, wherein the material comprises a titanium alloy.
3. The method of claim 1, wherein the pixel is a first pixel, the method further comprising: determining, by the processing circuitry, an alpha grain of each pixel of the plurality of pixels based on a position of each pixel and the first and second Euler angles (ϕ_1 , Φ) of each pixel; determining, by the processing circuitry, which alpha grains of the plurality of pixels may be from the same prior beta grain, and adjusting, by the processing circuitry, the third Euler angle (ϕ_2) of a second pixel of the plurality of pixels, wherein adjusting the third Euler angle (ϕ_2) of the first pixel and the second pixel is based on: the first and second pixels are from different alpha grains, the first and second pixels being from the same prior beta grain, and the first and second pixels being adjacent to each other.
4. The method of claim 3, wherein determining which alpha grains of the plurality of pixels may be from the same prior beta grain comprises: randomly selecting a pixel of the plurality of pixels as the first pixel, wherein the first pixel is from a first alpha grain; selecting the second pixel based on the second pixel being from a second alpha grain different from the first alpha grain and the second pixel being adjacent to the first pixel; and determining that a c-axis misorientation between the first and second pixels is one of about 0° , about 60° , or about 90° .
5. The method of claim 4, further comprising: assigning, by the processing circuitry, a random angle value to the third Euler angle (ϕ_2) of the first pixel; assigning, by the processing circuitry, a different angle value to the third Euler angle (ϕ_2) of the second pixel such that the c-axis misorientation between the first and second pixels is one of about 0° , about 60° , or about 90° ; assigning, by the processing circuitry, the random angle value to the third Euler angle (ϕ_2) of each pixel of the plurality of pixels that is from the first alpha grain; and assigning, by the processing circuitry, the different angle value to the third Euler angle (ϕ_2) of each pixel of the plurality of pixels that is from the second alpha.
6. The method of claim 5, further comprising: determining, by the processing circuitry, a first Burgers inverse transform (IT) solution to transform the first pixel to a first beta phase pixel of a beta phase data set and a second Burgers inverse transform (IT) solution to transform the second pixel to a second beta phase pixel of the beta phase data set, wherein determining the first and second Burgers IT solutions is based on reducing the total beta phase misorientation of the beta phase data set after transforming the first and second pixels by the first and second Burgers IT solutions.
7. The method of claim 6, wherein the first and second Burgers IT solutions are selected from a subset of less than all six of a set of six unique Burgers IT solutions that transform an alpha phase crystal to a beta phase crystal.
8. The method of claim 7, further comprising: determining, by the processing circuitry, a Burgers IT solution from the subset of less than all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution; and subsequently determining, by the processing circuitry, a Burgers IT solution from the subset including all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution.
9. The method of claim 8, further comprising: changing, by the processing circuitry, the value of the third Euler angle (ϕ_2) of the first pixel by at least one of 30° , 10.53° , or 5.16° ; and determining, by the processing circuitry, a Burgers IT solution from the subset including all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution.
10. The method of claim 9, wherein a blade of a gas turbine engine comprises the material, the method further comprising: determining, by the processing circuitry and subsequent to changing

the value of the third Euler angle (ϕ_2) of the first pixel, at least one microtexture region (MTR) based on the alpha phase data set; outputting, by the processing circuitry, an alert based on determining the at least one MTR; determining, by the processing circuitry, a status of the blade based on at least one of the alpha phase data set or the at least one MTR; and outputting, by the processing circuitry, the status of the blade.

11. A device comprising: a measurement device configured to measure a crystalline structure of a material and output an alpha phase data set indicative of the measurement of the crystalline structure; a memory; and one or more processors implemented in circuitry and in communication with the memory, the one or more processors configured to: receive the alpha phase data set, wherein the alpha phase data set comprises a plurality of pixels, wherein each pixel of the plurality of pixels includes a position, a first Euler angle (ϕ_1), a second Euler angle Φ , and a third Euler angle (ϕ_2), wherein the third Euler angle (ϕ_2) is missing or erroneous; adjust the third Euler angle (ϕ_2) of a pixel of the plurality of pixels; and store, based on adjusting the third Euler angle (ϕ_2) of the pixel reducing a total beta phase misorientation, the alpha phase data set.

12. The device of claim 11, wherein the material comprises a titanium alloy.

13. The device of claim 11, wherein the pixel is a first pixel, wherein the one or more processors are further configured to: determine an alpha grain of each pixel of the plurality of pixels based on a position of each pixel and the first and second Euler angles (ϕ_1 , Φ) of each pixel; determine which alpha grains of the plurality of pixels may be from the same prior beta grain, and adjust the third Euler angle (ϕ_2) of a second pixel of the plurality of pixels, wherein adjusting the third Euler angle (ϕ_2) of the first pixel and the second pixel is based on: the first and second pixels are from different alpha grains, the first and second pixels being from the same prior beta grain, and the first and second pixels being adjacent to each other.

14. The device of claim 13, wherein, to determine which alpha grains of the plurality of pixels may be from the same prior beta grain, the one or more processors are configured to: randomly select a pixel of the plurality of pixels as the first pixel, wherein the first pixel is from a first alpha grain; select the second pixel based on the second pixel being from a second alpha grain different from the first alpha grain and the second pixel being adjacent to the first pixel; and determine that a c-axis misorientation between the first and second pixels is one of about 0° , about 60° , or about 90° .

15. The device of claim 14, wherein the one or more processors are further configured to: assign a random angle value to the third Euler angle (ϕ_2) of the first pixel; assign a different angle value to the third Euler angle (ϕ_2) of the second pixel such that the c-axis misorientation between the first and second pixels is one of about 0° , about 60° , or about 90° ; assign the random angle value to the third Euler angle (ϕ_2) of each pixel of the plurality of pixels that is from the first alpha grain; and assign the different angle value to the third Euler angle (ϕ_2) of each pixel of the plurality of pixels that is from the second alpha.

16. The device of claim 15, wherein the one or more processors are further configured to: determine a first Burgers inverse transform (IT) solution to transform the first pixel to a first beta phase pixel of a beta phase data set and a second Burgers inverse transform (IT) solution to transform the second pixel to a second beta phase pixel of the beta phase data set, wherein determining the first and second Burgers IT solutions is based on reducing the total beta phase misorientation of the beta phase data set after transforming the first and second pixels by the first and second Burgers IT solutions.

17. The device of claim 16, wherein the first and second Burgers IT solutions are selected from a subset of less than all six of a set of six unique Burgers IT solutions that transform an alpha phase crystal to a beta phase crystal.

18. The device of claim 17, wherein the one or more processors are further configured to: determine a Burgers IT solution from the subset of less than all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution; and

subsequently determine a Burgers IT solution from the subset including all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution.

19. The device of claim 18, wherein a blade of a gas turbine engine comprises the material, wherein the one or more processors are further configured to: change the value of the third Euler angle (ϕ_2) of the first pixel by at least one of 30° , 10.53° , or 5.16° ; determine a Burgers IT solution from the subset including all six unique Burgers IT solutions for each pixel of the plurality of pixels based on reducing the total beta phase misorientation of the beta phase data set after transforming each pixel by the respective Burgers IT solution; determine, subsequent to changing the value of the third Euler angle (ϕ_2) of the first pixel, at least one microtexture region (MTR) based on the alpha phase data set; output and alert based on determining the at least one MTR; determine a status of the blade based on at least one of the alpha phase data set or the at least one MTR; and output the status of the blade.

20. A non-transitory computer-readable storage medium having stored thereon instructions that, when executed, configure a processor to: receive an alpha phase data set, wherein the alpha phase data set comprises a plurality of pixels, wherein each pixel of the plurality of pixels includes a position, a first Euler angle (ϕ_1), a second Euler angle Φ , and a third Euler angle (ϕ_2), wherein the third Euler angle (ϕ_2) is missing or erroneous; adjust the third Euler angle (ϕ_2) of a pixel of the plurality of pixels; and store, based on adjusting the third Euler angle (ϕ_2) of the pixel reducing a total beta phase misorientation, the alpha phase data set.
